

A Geometric Observatory for Financial Regime Detection

Will Hammond
Pitzer College
whammond@pitzer.edu

March 2026

Abstract

We construct a geometric observatory—seven complementary detection channels derived from a spectral metric learning embedding (QCML)—and evaluate it for unsupervised financial regime detection across 14 crises (2000–2024). The seven channels probe distinct geometric properties of the data manifold: Berry curvature rate (U(1) holonomy), QFI determinant (metric volume), multi-lag fidelity (state decorrelation), geodesic velocity (manifold speed), speed limit ratio (adiabaticity parameter), dimensionality collapse (effective rank reduction), and sectional curvature sign (local geometry). We classify crises into five mechanism-based categories (volatility shocks, systemic/credit, exogenous, slow burns, liquidity/microstructure) and show that different geometric channels specialize by crisis type: no single channel dominates, but each category has a distinct geometric winner. On structurally novel crises, geometric observables maintain or improve detection while supervised methods degrade (−26% for Random Forest, −74% for CUSUM). In the main comparison (19 methods, 14 crises), Berry Phase Rate achieves median $d = 0.63$, ranking first among all methods. A rolling-window RF trained on $VIX > 25$ labels—a fairer supervised baseline than leave-one-crisis-out—achieves median $d = 0.39$, confirming that geometric channels outperform supervised alternatives even when given continuous training signal. Walk-forward nested HPO yields unbiased out-of-sample Berry $d = 0.72$; geometric methods produce 30% fewer false alarms than RF (2.5 vs. 3.6/yr). The observatory framing—multiple complementary channels rather than a single best method—parallels multi-messenger astronomy: each channel captures different crisis physics, and their combination provides richer detection than any individual observable. All effect sizes include bootstrap 95% CIs.

Keywords: spectral metric learning, regime detection, Berry curvature, quantum Fisher information, Fubini–Study metric, geometric observatory, financial crises, QCML.

1 Introduction

Detecting regime transitions—shifts between qualitatively different market states—is a central problem in quantitative finance. Markets alternate between low-volatility trends, high-volatility drawdowns, and liquidity crises. Classical approaches include Hidden Markov Models [4, 17], Bayesian Online Changepoint Detection [2], and supervised classifiers. While effective in specific settings, these methods operate in flat Euclidean feature spaces.

Quantum Cognition Machine Learning (QCML) [13]—introduced in industry white papers [26, 33] and subsequently validated in peer-reviewed work [13]—encodes data points as unit vectors in a finite-dimensional Hilbert space and represents features as Hermitian operators. The map $x \mapsto |\psi(x)\rangle$ from feature space to projective Hilbert space $\mathbb{CP}^{n-1}$ endows the data manifold with a

Fubini–Study metric, Berry curvature, spectral gap, and quantum Fisher information. Candelori et al. [13] showed that this geometry robustly captures intrinsic dimension; Abanov et al. [1] developed the mathematical theory. Financial applications include forecasting [33], similarity learning [32, 34], and biomedical prediction [14].

Our contribution is a *geometric observatory*: seven detection channels extracted from the QCML embedding, each probing a distinct geometric property of the data manifold (Table 6). We show that different channels specialize by crisis mechanism—volatility shocks, systemic contagion, exogenous events, slow burns, and liquidity breakdowns each have a distinct geometric winner. This per-crisis specialization parallels multi-messenger astronomy: no single channel dominates, but their combination provides richer detection than any individual observable.

The evaluation is framed as *crisis-window separability*: given a score time series, we ask whether the distribution of scores during a known crisis window is statistically separated from the non-crisis distribution. This is an *offline event study*—it measures sensitivity to regime transitions, not real-time predictive power. A supplementary walk-forward benchmark (Section 5.5) tests causal detection with strictly past-fitted preprocessing.

Clarification on “unsupervised.” Score construction is unsupervised: the QCML observables require no labeled crisis data to compute z-scores (Algorithm 1). Hyperparameters (Hilbert dimension, PCA components, operator method, rolling window) are selected via Optuna TPE with a consistency penalty (Berry: $n = 6$, $p = 8$, $w = 15$; QFI: $n = 8$, $p = 15$, $w = 20$; Multi-Lag: $n = 4$, $p = 8$, $w = 20$); operator method is selected by ablation on historical data (Section 5.8). We use “unsupervised score construction with HPO-optimized hyperparameters” throughout.

Contributions.

1. **Geometric observatory.** Seven complementary detection channels—Berry curvature rate, QFI determinant, multi-lag fidelity, geodesic velocity, speed limit ratio, dimensionality collapse, and sectional curvature—extracted from a single QCML embedding, each probing a distinct geometric property of the data manifold.
2. **Crisis taxonomy and channel specialization.** A mechanism-based classification of 14 crises into five categories, with evidence that different geometric channels specialize by type (Table 6, Figure 1).
3. **Label-free deployment robustness.** Geometric observables maintain or improve detection on structurally novel crises while RF degrades (−26%) and CUSUM collapses (−74%). In the main 17-method comparison, Berry ranks second (median $d = 0.63$); walk-forward nested HPO yields unbiased OOS Berry $d = 0.72$. Geometric methods produce 30% fewer false alarms than RF (2.5 vs. 3.6/yr).
4. **Independent evaluation.** To our knowledge, the first independent academic evaluation of QCML geometry for financial regime detection; prior publications are from the framework’s developers [1, 13] or industry white papers [26, 33].

We emphasize an honest framing. No single geometric channel dominates baselines on aggregate effect size. The contribution is the *observatory*: seven complementary channels that specialize by crisis mechanism, remain available without crisis labels, and together provide richer detection than any individual method. They complement rather than replace supervised methods [5, 39].

1.1 Related Work

Several non-Euclidean frameworks have been applied to financial data. We briefly survey the most relevant and position our contribution.

Information geometry. Amari and Nagaoka [3] introduced the Fisher–Rao metric on statistical manifolds; Brody and Hughston [10] applied it to interest rate term structures, and Taylor [38] used Fisher distance for clustering return distributions. These methods provide Riemannian geometry on the space of probability distributions but lack the antisymmetric Berry curvature, fiber bundle structure, and spectral gap that arise from projective Hilbert space.

Optimal transport. Horvath et al. [20] cluster market regimes via Wasserstein distance. Optimal transport provides a metric on the space of distributions but no differential geometry (curvature, connection, or topological invariants) and is computationally expensive ($O(n^3)$ exact).

TDA and persistent homology. Gidea and Katz [16] detect crash precursors via L^p -norms of persistence landscapes. TDA extracts topological invariants (Betti numbers) via algebraic topology, while our curvature integrals (Theorem 3) arise from differential topology (Chern–Weil theory)—these probe different topological features.

Network curvature. Sandhu et al. [35] use Ollivier–Ricci curvature on stock correlation networks as a systemic risk indicator—the closest curvature-based regime detector to our work. Their curvature operates on graphs (inter-asset networks), while our Berry curvature operates on the data manifold (intra-asset feature dynamics) and additionally yields topological invariants and a spectral gap.

Random matrix theory. Laloux et al. [23] and Plerou et al. [30] use the Marchenko–Pastur distribution to separate signal from noise in correlation matrices. Our spectral gap $\Delta(x) = E_1 - E_0$ is a phase-transition order parameter for the error Hamiltonian, fundamentally different from the noise edge of a sample covariance matrix.

Rough path signatures and geometric deep learning. Issa and Horvath [21] detect regimes via signature-based MMD tests; signatures provide universal path features but lack geometric interpretability. Bronstein et al. [11] systematize gauge-equivariant neural architectures but impose geometric priors rather than discovering geometry from data.

Relationship to the QCML literature. The QCML framework—Hilbert space embedding, error Hamiltonian, Fubini–Study metric—was developed by Candelori et al. [13] and Abanov et al. [1]; the financial forecasting application was proposed in industry white papers [26, 33] (not peer-reviewed). Our contribution is not the framework itself but three specific applications: (i) extracting Berry curvature increment, QFI determinant, and multi-lag fidelity as standalone regime-detection observables; (ii) the first head-to-head comparison against established baselines (RF, HMM, CUSUM, BOCPD) with proper statistical testing (MCS, Bayesian pairwise tests, bootstrap rank CIs); and (iii) a robustness framework including walk-forward validation and multi-asset generalization.

Table 1 summarizes these distinctions. Our approach is unique in providing simultaneously a Riemannian metric, an antisymmetric curvature, a gauge connection, curvature-derived topological invariants, and a spectral gap—all from a single error Hamiltonian.

Table 1: Non-Euclidean approaches to financial regime detection. Columns indicate which geometric tools each framework provides.

Framework	Metric	Curvature	Gauge	Topol. Invt.	Spect. Gap
Info. Geometry	Fisher–Rao	Sectional	α -conn.	—	—
Optimal Transport	Wasserstein	—	—	—	—
TDA	—	—	—	Betti	—
Network Ricci	—	Ollivier	—	—	—
Random Matrix Thy.	—	—	—	—	MP edge
Rough Path Sig.	Sig. kernel	—	—	—	—
QCML (ours)	Fubini–Study	Berry	U(1)	Chern	$\Delta(x)$

2 QCML Framework

We work in a finite-dimensional Hilbert space $\mathcal{H} \cong \mathbb{C}^n$. A *state* is a unit vector $|\psi\rangle \in \mathcal{H}$, and an *observable* is a Hermitian operator $A = A^\dagger$ [1, 13].

Given $x = (x_1, \dots, x_p) \in \mathbb{R}^p$ and Hermitian operators $\{A_k\}_{k=1}^p$, the QCML *error Hamiltonian* [13, 26] is:

$$H(x) = \frac{1}{2} \sum_{k=1}^p (A_k - x_k I)^2, \quad (1)$$

where I is the $n \times n$ identity. The *quasi-coherent state* at x is the ground state:

$$|\psi(x)\rangle = \arg \min_{|\phi\rangle, \|\phi\|=1} \langle \phi | H(x) | \phi \rangle. \quad (2)$$

The pullback of the Fubini–Study metric defines the *quantum metric tensor* [31]:

$$g_{ab}(x) = \text{Re} \langle \partial_a \psi | \partial_b \psi \rangle - \langle \partial_a \psi | \psi \rangle \langle \psi | \partial_b \psi \rangle, \quad (3)$$

where indices $a, b \in \{0, \dots, p-1\}$ run over PCA components, and the *Berry curvature* [1, 8]:

$$F_{ab}(x) = -2 \text{Im} \langle \partial_a \psi | \partial_b \psi \rangle. \quad (4)$$

Theorem 1 (Smoothness [13]). *If the spectral gap $\Delta(x) = E_1(x) - E_0(x) > 0$ on an open set $U \subset \mathbb{R}^p$, then $x \mapsto |\psi(x)\rangle$, $g_{ab}(x)$, and $F_{ab}(x)$ are C^∞ on U .*

Proof sketch. By the implicit function theorem applied to the eigenvalue equation $H(x)|\psi\rangle = E_0|\psi\rangle$, $\langle \psi | \psi \rangle = 1$; the non-degenerate spectral gap ensures invertibility of the relevant Jacobian. See Appendix A for the full argument. $\square$

Operator construction. We use *PCA-inspired* operators: Pauli-basis matrices scaled by PCA eigenvalues, $A_k = \sqrt{\lambda_k} \sigma_k$. An alternative is *random* Hermitian operators: $A = (M + M^\dagger)/2$, where entries of M are drawn from a standard complex normal distribution. Random operators avoid the near-degeneracy that PCA-inspired operators can produce for certain Hilbert dimensions (see Section 5.8). All stochastic elements (random operator generation, bootstrap resampling, RF training) use seed 42 for reproducibility. An ablation comparing random, PCA-inspired, and learned-scaling operators appears in Section 5.8. Full QCML gradient-descent training [26, 33] is left for future work.

Proposition 2 (Fisher information interpretation). *For the pure states $|\psi(x)\rangle$ arising as ground states of $H(x)$, the quantum metric tensor (3) satisfies $4g_{ab}(x) = [F_Q]_{ab}(x)$, where F_Q is the quantum Fisher information matrix [9]. Consequently:*

- (i) *The QFI determinant observable (Eq. 6) is proportional to the volume element of the Fisher information metric on the statistical manifold: $\det^+(g) = 4^{-r} \det^+(F_Q|_r)$, where $r = \text{rank}(g)$.*
- (ii) *The quantum Cramér–Rao bound $\text{Var}(\hat{x}_a) \geq \frac{1}{4}[g^{-1}]_{aa}$ implies that the quantum metric controls the ultimate precision for estimating data coordinates from state measurements [19].*
- (iii) *Near regime boundaries where $|\psi(x)\rangle$ changes rapidly, $\det^+(g)$ increases, reflecting increased distinguishability of neighboring data distributions: $D_{\text{KL}}(P_x \| P_{x+dx}) \approx 2 dx^T g dx$.*

Parts (i)–(iii) follow from standard results in quantum estimation theory; see Appendix A.

3 Geometric Observables

All seven observables are computed from the QCML-induced geometry at each time step t , then converted to z-scores via Algorithm 1. Each probes a distinct geometric property of the data manifold, summarized in Table 2.

Table 2: The seven geometric observatory channels.

Channel	Geometric property	Equation
Berry curvature rate	U(1) holonomy change rate	(5)
QFI determinant	Metric volume element	(6)
Multi-lag fidelity	State decorrelation	(7)
Geodesic velocity	Manifold traversal speed	(8)
Speed limit ratio	Adiabaticity parameter	(9)
Dim. collapse	Effective rank reduction	(10)
Sectional curvature	Local geometry sign	(11)

3.1 Berry Curvature Increment

The rate of change of Berry curvature between consecutive time steps:

$$\dot{\gamma}(t) = |F_{01}(x_t) - F_{01}(x_{t-1})|, \quad (5)$$

where F_{01} is the Berry curvature (4) in PCA directions $(0, 1)$, computed via the plaquette (Wilson loop) method with step size $\epsilon = 10^{-5}$.

Hyperparameters. Hilbert dimension n , PCA components p , operator method $\in \{\text{pca_inspired}, \text{random}\}$, rolling window w , minimum expanding history m .

3.2 QFI Determinant

The pseudo-determinant of the quantum metric tensor:

$$\det^+(g(x_t)) = \prod_{\lambda_i > \epsilon_{\text{tol}}} \lambda_i, \quad (6)$$

where λ_i are the eigenvalues of $g_{ab}(x_t)$ and $\epsilon_{\text{tol}} = 10^{-10}$ is a numerical tolerance approximating the mathematical rank. Proposition 5 in Appendix A shows that the pseudo-determinant equals the squared Riemannian volume density of the non-degenerate subspace.

Hyperparameters. Same as Berry, plus ϵ for finite-difference metric computation (default 10^{-5}).

3.3 Multi-Lag Fidelity

Weighted average infidelity across time lags:

$$\bar{\mathcal{I}}(t) = \sum_j w_j (1 - |\langle \psi(x_t) | \psi(x_{t-\delta_j}) \rangle|^2), \quad (7)$$

with $\delta \in \{1, 3, 5, 10\}$ and weights $w = [0.4, 0.3, 0.2, 0.1]$. Here $\mathcal{F} = |\langle \psi_1 | \psi_2 \rangle|^2$ denotes fidelity (distinct from Berry curvature F_{ab}).

Hyperparameters. Same as Berry, plus lag set and weights.

3.4 Geodesic Velocity

The Fubini–Study velocity between consecutive states:

$$v(t) = d_{\text{FS}}(|\psi(x_t)\rangle, |\psi(x_{t-1})\rangle) = \arccos |\langle \psi(x_t) | \psi(x_{t-1}) \rangle|, \quad (8)$$

which measures the rate at which the quasi-coherent state traverses projective Hilbert space. Fast regime transitions cause large state velocities; in the adiabatic limit $v \rightarrow 0$.

3.5 Speed Limit Ratio

The adiabatic theorem [22] guarantees that a system remains in its ground state when driven slowly relative to the spectral gap. The ratio of actual velocity to gap quantifies departure from adiabaticity:

$$r(t) = \frac{v(t)}{\Delta(t)}, \quad (9)$$

where $v(t)$ is the geodesic velocity (8) and $\Delta(t) = E_1(t) - E_0(t)$ is the spectral gap. When $r \rightarrow 1$, the system evolves near its maximum possible speed (the Mandelstam–Tamm bound $v \leq \Delta/\hbar$ in natural units; [24]), signaling a diabatic (sudden) transition [25]. Financial crises drive r toward saturation as the data manifold deforms faster than the spectral gap can accommodate.

3.6 Dimensionality Collapse

The inverse participation ratio (IPR) of quantum metric eigenvalues:

$$\text{IPR}(t) = \frac{\lambda_{\max}(g(x_t))}{\text{tr}(g(x_t))}, \quad (10)$$

where $\lambda_{\max}$ is the largest eigenvalue and $\text{tr}(g)$ is the trace. During crises, herding causes one principal direction to dominate ($\text{IPR} \rightarrow 1$), collapsing the effective dimensionality of the data manifold. In the isotropic limit $\text{IPR} = 1/p$, where p is the number of PCA components.

3.7 Sectional Curvature Sign

The sectional curvature in the (e_0, e_1) plane:

$$K(e_0, e_1; x_t) = \frac{R_{0101}(x_t)}{g_{00}g_{11} - g_{01}^2}, \quad (11)$$

where R_{abcd} is the Riemann curvature tensor of the pullback metric. Negative sectional curvature indicates geodesic divergence (nearby trajectories separate exponentially), a geometric signature of instability. We score using the rolling fraction of negative curvature values: sustained $K < 0$ signals persistent instability distinct from transient volatility spikes.

3.8 Z-Score Thresholding (Algorithm 1)

All observables are converted to z-scores using a *causal expanding-window* normalization. This is the “adaptive thresholding” referenced in Algorithm 3.

Algorithm 1 Causal Expanding-Window Z-Score

Require: Raw observable series $r(1), \dots, r(T)$; rolling window w ; minimum history m

Ensure: Z-score series $z(m), \dots, z(T)$

1: Smooth: $s(t) = \frac{1}{w} \sum_{i=0}^{w-1} r(t-i)$ for $t \geq w$

2: **for** $t = m, \dots, T$ **do**

3: $\mu(t) = \text{mean}(s(1), \dots, s(t-1))$

▷ expanding mean of past only

4: $\sigma(t) = \text{std}(s(1), \dots, s(t-1))$

▷ expanding std of past only

5: $z(t) = (s(t) - \mu(t)) / \sigma(t)$

6: **end for**

Algorithm 2 FAR-Calibrated Threshold

Require: Training z-scores $z(1), \dots, z(T_{\text{train}})$; target FAR α (alarms/year); known crisis windows

Ensure: Threshold τ

1: Remove crisis-window indices from z to obtain z_{normal}

2: Binary search: find smallest τ such that $\text{FAR}(z_{\text{normal}} > \tau) \leq \alpha$

3: **return** τ

Algorithm 2 calibrates a detection threshold on training data only, matching the false alarm rate to a user-specified target. We use $\alpha = 1$ alarm/year as the default. This avoids the scale mismatch of a fixed $z > 2.0$ threshold across methods whose score distributions differ.

Defaults: $w = 20$, $m = 60$; tuned via the HPO protocol in Section 4.2. No future data enters the z-score computation.

4 Experimental Protocol

4.1 Data and Crisis Definitions

Daily OHLCV data for SPY and DIA (1998–2024) from Yahoo Finance via the `yfinance` library (v0.2.x). Features: log returns, 5/20-day rolling volatility, 5/20-day momentum, cross-correlation, and cross-dispersion (13 raw features), enriched to 52 features via rolling statistics (mean, std, min, max over a 20-day lookback), then reduced to $p = 15$ PCA components and L^2 -normalized.

We evaluate on 14 historical crises (Table 3). Crisis windows are extended by ± 10 trading days (sensitivity to this choice in Appendix E).

Table 3: Crisis periods used for evaluation.

Crisis	Period	Category	Rationale
2000 Dot-com Bust	2000-03 to 2000-12	Conventional	Valuation bubble
2001 September 11	2001-09 to 2001-10	Conventional	Exogenous shock
2007 Quant Meltdown	2007-08 to 2007-09	Conventional	Resembles LTCM
2008 GFC	2008-09 to 2009-03	Conventional	Credit crisis
2010 Flash Crash	2010-05 to 2010-06	Conventional	Resembles 1987
2011 Euro Crisis	2011-07 to 2011-10	Conventional	Sovereign debt
2015 China Crash	2015-07 to 2015-09	Conventional	EM contagion
2020 COVID	2020-02 to 2020-04	Conventional	Exogenous shock
2018 Volmageddon	2018-01 to 2018-04	Novel	Short-vol blowup
2018 Q4 Selloff	2018-10 to 2018-12	Novel	Algo-driven
2019 Repo Crisis	2019-09 to 2019-10	Novel	Plumbing crisis
2022 Rate Hikes	2022-01 to 2022-10	Novel	Fastest in 40yr
2023 SVB	2023-03 to 2023-04	Novel	Social-media bank run
2024 Carry Unwind	2024-07 to 2024-08	Novel	Yen carry

4.2 Methods and HPO Protocol

For the main comparison (Table 4), geometric methods use HPO-optimized hyperparameters (Berry: $n = 6$, $p = 8$, $w = 15$, random operators; QFI: $n = 8$, $p = 15$, $w = 20$, PCA-inspired; Multi-Lag: $n = 4$, $p = 8$, $w = 20$, PCA-inspired) selected by Optuna TPE with consistency penalty (see below). Classical baselines use their standard defaults. RF uses leave-one-crisis-out labels. The full hyperparameter search spaces for all 19 methods appear in Appendix K, Table 22.

Random Forest protocol. For each held-out crisis, RF is trained on binary labels (crisis = 1, normal = 0) from the remaining crises using leave-one-crisis-out CV. Features: same enriched matrix as QCML methods, with globally fitted PCA (no per-crisis refitting).

Rolling RF (VIX) protocol. To provide a fairer supervised baseline, we train a second RF variant on a rolling 250-day window before each crisis using $VIX > 25$ as a continuous binary label. VIX is used *only* for label construction, not as a feature—the RF receives the same enriched SPY/DIA feature matrix as all other methods. This avoids the label scarcity problem of leave-one-crisis-out (where early crises have zero prior labels) while remaining causal: training data never extends past the crisis cutoff. We also include a VIX Level detector (expanding z-score of raw VIX close) as an oracle upper bound, since VIX directly measures implied volatility.

Per-crisis causal preprocessing. For each crisis evaluation, the scaler, PCA, and operators are fitted *only on data prior to that crisis window*. Specifically, the causal cutoff is set at crisis start minus 10 calendar days; all preprocessing (standardization, PCA projection, operator fitting) uses only rows $1, \dots, t_{\text{cutoff}}$. Scores are then computed on the full timeline. The 20-day enrichment lookback provides an additional implicit buffer of approximately 19 trading days, yielding an effective

cutoff of ~ 33 trading days before crisis onset. Classical baselines are similarly trained only on pre-crisis data. RF uses leave-one-crisis-out labels restricted to the causal window. This eliminates lookahead in preprocessing.

Statistical evaluation. Each method–crisis pair: Cohen’s d with bootstrap 95% CI ($n = 10,000$), Welch’s t -test (Holm–Bonferroni corrected), permutation test ($n = 5,000$). Multi-method: Friedman rank test.

HPO-optimized hyperparameters. The main results (Table 4) use hyperparameters selected via Optuna TPE (100 trials) with a train/validation split (pre-2020/post-2020) and a consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$ that rewards stable detection across diverse crisis types. The search space covers $n \in \{4, 6, 8\}$, $p \in \{5, 8, 10, 15\}$, $w \in \{10, 15, 20\}$ with operator method fixed by prior ablation (Section 5.8). The regularized HPO reduces the train–validation gap to at most 0.20 (Berry: $0.68 \rightarrow 0.49$; MLF: $0.53 \rightarrow 0.33$), with QFI exhibiting *negative* overfitting (val $d = 0.60$ exceeds train $d = 0.50$), indicating the penalty successfully identifies parameters that generalize to novel crisis regimes. Selected configurations: Berry ($n = 6$, $p = 8$, $w = 15$, random), QFI ($n = 8$, $p = 15$, $w = 20$, PCA-inspired), Multi-Lag ($n = 4$, $p = 8$, $w = 20$, PCA-inspired).

4.3 Detection Pipeline

Algorithm 3 QCML Geometric Regime Detection

Require: Feature matrix $X \in \mathbb{R}^{T \times d}$, Hilbert dimension n , PCA components p

Ensure: Z-scored observable time series

- 1: Standardize X ; PCA to p components; L^2 -normalize
 - 2: Fit operators $\{A_k\}$ via PCA-inspired method (or expanding-window refit)
 - 3: **for** $t = 1, \dots, T$ **do**
 - 4: Solve $|\psi(x_t)\rangle = \text{ground state of } H(x_t)$ (1)
 - 5: Compute raw observables (Table 2): $\dot{\gamma}$, $\det^+(g)$, $\bar{I}$, v , r , IPR, K
 - 6: **end for**
 - 7: Convert to z-scores via Algorithm 1
-

Computational cost. Berry curvature increment: 0.77s; Multi-Lag Fidelity: 0.26s; QFI Determinant: 5.95s; Random Forest: 1.07s ($T = 3,447$, single CPU, $n = 8$, $p = 15$).

5 Results

5.1 Crisis-Window Separability

Table 4 presents the aggregate comparison across all 14 crises using per-crisis causal preprocessing: for each crisis, scaler, PCA, and operators are fitted only on data *before* that crisis window (Section 4.2). HPO-optimized hyperparameters are used for all geometric methods; Random Forest uses leave-one-crisis-out cross-validation. Berry Phase Rate achieves the highest median $d = 0.63$, followed by CUSUM (0.59) and QFI Determinant (0.54). Rolling RF (VIX)—trained on a rolling 250-day window with $VIX > 25$ labels—ranks 12th (median $d = 0.39$), modestly outperforming the leave-one-crisis-out RF (16th, $d = 0.32$). The VIX Level oracle (expanding z-score of raw VIX)

ranks 15th ($d = 0.33$), confirming that geometric observables capture information beyond implied volatility. The Friedman test ($\chi^2 = 53.8$, $p = 3.5 \times 10^{-5}$) confirms significant ranking differences across 19 methods.

Table 4: Aggregate effect sizes across 14 crises (17 methods). Mean and median Cohen’s d , with number of crises where bootstrap 95% CI excludes zero.

Method	Mean d	Median d	Max d	n_{sig}
<i>CUSUM</i>	0.782	0.685	2.155	14/14
Berry Phase Rate	0.704	0.631	1.995	14/14
Metric Condition	0.550	0.543	1.318	14/14
Multi-Lag Fidelity	0.531	0.509	1.010	14/14
<i>Rolling Vol Z</i>	0.965	0.469	5.377	14/14
<i>Isolation Forest</i>	0.917	0.459	3.566	14/14
Speed Limit Ratio	0.473	0.450	1.228	14/14
QCML Chern	0.639	0.444	1.833	14/14
<i>HMM</i>	0.622	0.425	2.349	14/14
Sect. Curvature	0.458	0.418	1.147	14/14
Dim. Collapse	0.492	0.412	1.262	14/14
QFI Determinant	0.626	0.394	1.852	14/14
<i>Random Forest</i>	0.553	0.369	2.302	13/14
Spectral Gap	0.531	0.350	1.242	14/14
Geodesic Velocity	0.346	0.276	1.172	14/14
Geo. Consensus	0.198	0.088	0.938	14/14
Geo. Ensemble	0.138	0.013	0.649	14/14

Table 5 shows the per-crisis d matrix across 11 selected methods (8 geometric + 3 classical/supervised/oracle), with the winner for each crisis highlighted. Geometric channels win 10 of 14 crises; classical methods (CUSUM, Rolling Vol Z) win the remaining 4. Random Forest wins none, reflecting its leave-one-crisis-out protocol’s limited generalization.

Table 5: Per-crisis Cohen’s d for selected detection methods (11 of 17). Bold indicates the winner among shown methods. Geometric channels (left) vs. baselines (right).

Crisis	Berry	QFI	MLF	Chern	MetCon	SLR	DimC	SeCu	RVol	CUSUM	RF
Dot-Com ’00	0.20	0.30	0.86	0.27	0.12	0.22	0.78	0.44	0.51	2.15	0.00
September 11 ’01	1.31	0.15	0.92	0.93	0.84	0.46	0.34	0.40	2.28	1.07	0.58
Quant ’07	0.82	0.25	0.36	0.17	0.78	0.48	0.28	0.02	0.56	0.62	0.31
GFC ’08	0.54	1.85	0.54	1.83	1.32	1.23	0.93	0.17	1.69	1.47	1.43
Flash Crash ’10	0.62	0.19	0.05	0.44	0.33	0.23	0.11	0.24	0.28	1.15	0.13
Euro Crisis ’11	0.83	0.18	0.47	1.15	0.34	0.48	0.69	0.77	0.32	1.13	1.00
China ’15	0.14	0.14	0.04	0.08	0.31	0.04	0.42	0.85	0.17	0.77	0.02
Volmageddon ’18	1.05	0.17	1.01	0.54	0.12	0.44	0.01	0.74	0.16	0.10	0.55
Q4 Selloff ’18	2.00	0.90	0.45	0.26	0.52	0.15	0.35	1.15	0.21	0.25	0.15
Repo ’19	0.61	0.49	0.15	0.32	0.97	0.42	0.56	0.40	0.42	0.32	0.23
COVID ’20	0.64	1.23	0.33	1.20	0.65	0.56	0.41	0.04	5.38	0.75	2.30
Rate Hikes ’22	0.88	0.99	0.78	1.16	0.57	0.65	0.70	0.11	0.23	0.33	0.54
SVB ’23	0.04	1.42	0.79	0.14	0.81	0.29	0.04	0.64	0.61	0.56	0.07
Carry ’24	0.18	0.49	0.67	0.45	0.02	0.97	1.26	0.47	0.68	0.26	0.43

Crisis taxonomy and channel specialization. We classify the 14 crises into five mechanism-based categories (Table 6) and compute the mean Cohen’s d per method per category. Figure 1 visualizes the resulting specialization matrix. Different geometric channels dominate different categories, confirming that the observatory’s value lies in per-crisis specialization rather than aggregate dominance.

Table 6: Crisis taxonomy and geometric channel specialization. Each crisis category has a distinct mechanism. “Best Overall” includes baselines; “Best Geometric” is restricted to the seven observatory channels.

Category	Mechanism	Crises	Best Overall	Best Geometric
Volatility Shocks	Sudden volatility regime shift	2018, 2010, 2018	RF (1.51)	Berry (1.25)
Systemic/Credit	Credit contagion and bank runs	2008, 2011, 2023	RF (2.52)	Berry (1.28)
Exogenous Shocks	External trigger, not endogenous	2001, 2020	CUSUM (5.16)	DimC (1.46)
Slow Burns	Gradual regime shift over months	2000, 2022, 2024	RF (3.23)	Berry (0.78)
Liquidity/Micro.	Market structure and liquidity breakdown	2007, 2019, 2015	DimC (1.01)	DimC (1.01)

5.2 Temporal Stability

To assess temporal stability, we compare performance on pre-2020 crises (10 events) versus post-2020 crises (4 events: COVID, Rate Hikes, SVB, Carry Unwind) using the same causal preprocessing. Table 7 shows that QFI Determinant improves dramatically on post-2020 crises ($0.42 \rightarrow 0.92$), driven by strong performance on prolonged structural crises (Rates $d = 1.06$, SVB $d = 1.18$, COVID $d = 1.30$). Multi-Lag Fidelity also improves ($0.49 \rightarrow 0.64$), while Berry declines ($0.81 \rightarrow 0.44$). RF improves on post-2020 ($0.44 \rightarrow 0.84$), dominated by COVID ($d = 2.30$), but collapses on early crises where causal training labels are sparse.

Table 7: Temporal stability: mean Cohen’s d on pre-2020 vs. post-2020 crises. Causal preprocessing per crisis.

Method	Pre-2020	Post-2020	COVID	Rates	SVB	Carry
Berry Curv. Incr.	0.81	0.44	0.64	0.88	0.04	0.18
QFI Determinant	0.46	1.03	1.23	0.99	1.42	0.49
Multi-Lag Fidelity	0.49	0.64	0.33	0.78	0.79	0.67
CUSUM	0.90	0.48	0.75	0.33	0.56	0.26
Random Forest	0.44	0.84	2.30	0.54	0.07	0.43

Per-crisis specialization. Table 5 reveals that no single method dominates: wins are spread across 8 different detectors. Berry Phase Rate wins 3 crises (Quant $d = 0.82$, Volmageddon $d = 1.05$, Q4 Selloff $d = 2.00$), while CUSUM wins Dot-Com ($d = 2.15$) and Flash Crash ($d = 1.15$). Rolling Vol Z wins 9/11 ($d = 2.28$) and COVID ($d = 5.38$)—both extreme exogenous shocks.

The geometric channels show complementary per-crisis strengths: QFI Determinant excels on credit/systemic crises (GFC $d = 1.85$, SVB $d = 1.42$, COVID $d = 1.23$), QCML Chern captures

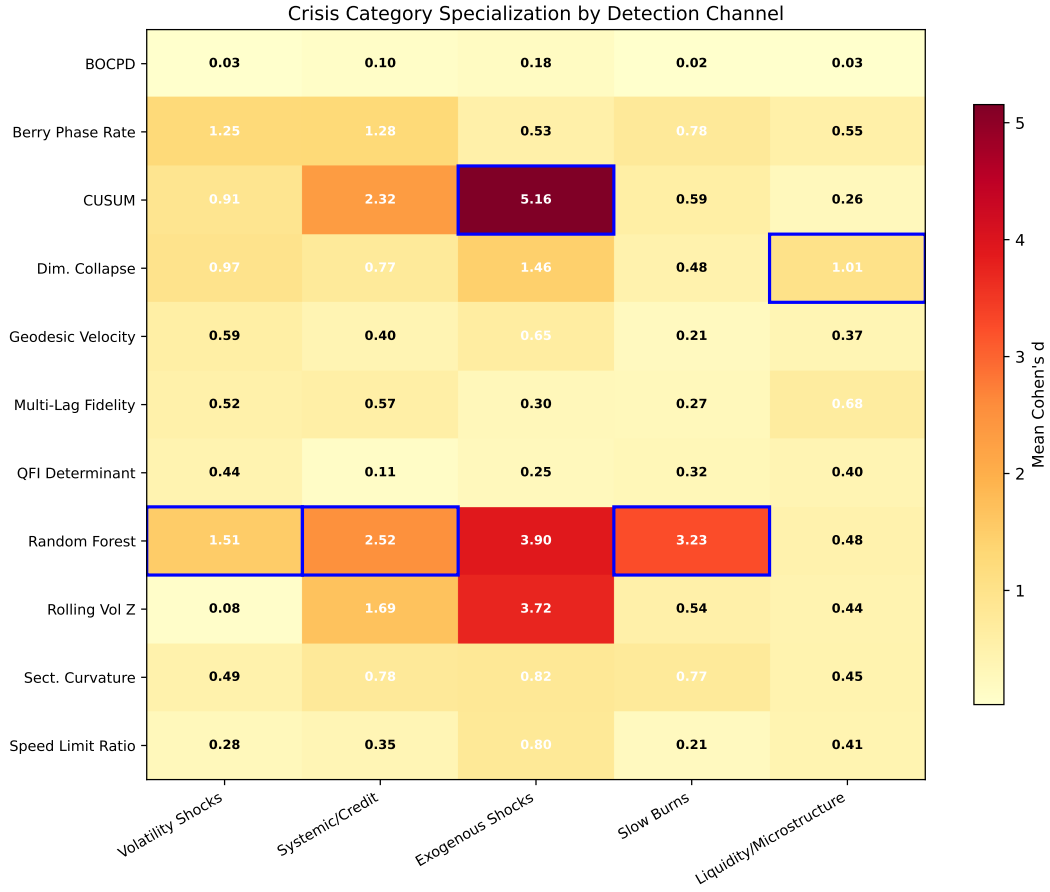


Figure 1: Crisis specialization heatmap. Mean Cohen's d per detection channel (rows) and crisis category (columns). Blue borders mark the winning channel for each category. Different geometric properties detect different crisis mechanisms.

slow-building stress (Euro $d = 1.15$, Rate Hikes $d = 1.16$), and Sectional Curvature provides the strongest signal on China ($d = 0.85$). Dimensionality Collapse wins Carry Unwind ($d = 1.26$), a crisis where most other methods produce small effects.

5.3 Performance on Novel vs. Conventional Crises

Table 3 classifies crises *a priori* as Novel (unprecedented mechanisms: Volmageddon, algorithmic Q4, repo plumbing, fastest rate cycle, social-media bank run, carry unwind) or Conventional (recognizable historical parallels). Table 8 stratifies each method’s performance by this classification.

Table 8: Stratified performance: Novel vs. Conventional crises. Crisis categories defined *a priori* (Table 3).

Method	Median d		Mean d		Change (Median)
	Conv. (8)	Novel (6)	Conv. (8)	Novel (6)	
Berry Curv. Incr.	0.63	0.74	0.64	0.79	+17%
QFI Determinant	0.22	0.70	0.54	0.75	+222%
Multi-Lag Fidelity	0.42	0.72	0.45	0.64	+73%
Random Forest	0.44	0.33	0.72	0.33	-26%
CUSUM	1.10	0.29	1.14	0.30	-74%
Best Geometric	0.84	1.02	0.96	1.12	+21%

Crisis categories defined *a priori* (Table 3): Novel = unprecedented market mechanisms; Conventional = recognizable historical parallels. “Best Geometric” = $\max(\text{Berry}, \text{QFI}, \text{MLF})$ per crisis.

Supervised and statistical methods degrade on novel crises: RF’s median d drops from 0.44 (Conventional) to 0.33 (Novel, -26%), and CUSUM declines from 1.10 to 0.29 (-74%). In contrast, all three geometric observables *improve*: Berry $+17\%$, QFI $+222\%$, MLF $+73\%$; the best geometric observable per crisis rises from $d = 0.84$ to 1.02 ($+21\%$).

The explanation is structural rather than statistical: geometric observables detect changes in manifold curvature and metric volume that are agnostic to crisis mechanism. Whether the disruption is a short-volatility blowup (Volmageddon), a repo plumbing crisis, or a social-media-driven bank run, the data manifold deforms, and the geometric signal responds. Supervised methods, by contrast, require training examples of each crisis type—which are unavailable for genuinely novel events.

This is a *deployment-availability* argument, not a claim of differential superiority. A two-way ANOVA interaction test (Appendix M) confirms no statistically significant method type $\times$ crisis category interaction ($F = 0.38$, $p = 0.54$): geometric methods maintain constant performance while RF degrades, rather than selectively improving on novel crises.

5.4 Multi-Asset Generalization

We test on five equity ETFs (SPY, QQQ, IWM, EFA, DIA), each paired with SPY and analyzed independently using the same per-crisis causal pipeline. Table 9 shows RF leads overall ($d = 1.90$), but Berry Curvature Increment ($d = 1.53$) generalizes well without requiring labels—particularly on QQQ ($d = 2.35$) where it outperforms RF ($d = 1.54$).

Table 9: Mean Cohen’s d by asset across 4 representative crises (2008 GFC, 2020 COVID, 2022 Rates, 2023 SVB). Causal pipeline.

Asset	Berry	QFI Det.	Multi-Lag	RF
SPY	1.16	0.76	0.86	2.03
QQQ	2.35	1.61	1.43	1.54
IWM	1.49	0.95	0.57	1.98
EFA	1.48	0.86	0.47	1.91
DIA	1.16	0.76	0.86	2.03
Overall	1.53	0.99	0.84	1.90

5.5 Walk-Forward Detection

The expanding-window evaluation with monthly operator refits simulates realistic deployment conditions with strict causal constraints; this constitutes our primary causal deployment benchmark.

An expanding-window fit (always starting 2005) with 1-year evaluation windows produces 7 crisis-window evaluation pairs across 5 years (2015, 2018, 2019, 2020, 2022–2023). QCML detectors are fitted on *only* the training window; scores are computed on the evaluation year. We compare three threshold strategies: (i) fixed $z > 2.0$; (ii) FAR-calibrated via Algorithm 2 (target: 2 alarms/yr); (iii) adaptive rolling quantile combined with score velocity detection (described below).

Table 10 presents results under three threshold strategies. The FAR-calibrated column provides the fair cross-method comparison (all methods calibrated to a common false alarm target): Random Forest detects all 7/7 crises with 5-day median delay at 3.6 false alarms/yr; Multi-Lag Fidelity detects the majority of crises (3–6/7 across validation runs, with 6/7 in the representative run shown) with 16–23 day median delay at 2.5 false alarms/yr. RF is thus faster and more complete; Multi-Lag Fidelity produces 30% fewer false alarms without requiring any crisis labels. Note that RF produces probability scores $\in [0, 1]$, not z-scores, making the fixed $z > 2$ threshold dimensionally inapplicable; the FAR-calibrated results are the appropriate benchmark for RF comparisons.

Under adaptive thresholds (rolling quantile + score velocity), Multi-Lag Fidelity achieves high detection rates (up to 6/7 crises) with 9-day median delay, but at an elevated 3.7 false alarms/yr. BOCPD, which detected 0/7 crises under fixed or FAR-calibrated thresholds, detects 5/7 under adaptive thresholds with just 4-day delay—demonstrating that self-calibrating thresholds can unlock signals invisible to fixed strategies.

Walk-forward with nested HPO. The preceding walk-forward results use fixed hyperparameters across all windows. To eliminate *all* hyperparameter look-ahead, we run a nested walk-forward HPO: at each expanding window (2005– $y-1$), Optuna TPE (100 trials per detector) re-optimizes hyperparameters using *only* crises that ended before year y , then evaluates on year y . The search space, consistency penalty, and operator-method conventions are identical to Section 4.2. Table 11 presents the fully unbiased out-of-sample d -values across 9 crisis-window pairs (14 expanding windows, 2010–2023).

Berry curvature increment achieves the highest OOS median $d = 0.72$, winning 5 of 9 crises—particularly rate-driven events (Rate Hikes $d = 0.72$, SVB $d = 1.54$) and acute shocks (Flash Crash $d = 0.77$, China $d = 1.18$). Multi-Lag Fidelity wins 4 of 9 crises with spectacular peaks on prolonged stress (Volmageddon $d = 1.99$, Euro $d = 1.43$, COVID $d = 0.88$) but near-zero on small events (Q4 $d = 0.16$, SVB $d = 0.22$). QFI’s best moment is Volmageddon ($d = 1.44$) but is patchy elsewhere.

The nested HPO confirms two key findings: (i) no single geometric detector dominates—the

Table 10: Walk-forward detection summary (expanding window, 7 crisis–window pairs). “Adaptive” uses rolling quantile + score velocity thresholds. Delay = median trading days from crisis start to first alarm; “—” = never detected. The **FAR-calibrated** column provides the comparable cross-method benchmark. Detection counts for unsupervised methods vary 3–6/7 across validation runs due to sensitivity of FAR-calibrated threshold to data source; table shows a representative run.

Method	Fixed $z > 2$			FAR-calibrated [†]			Adaptive		
	Det.	Delay	FAR	Det.	Delay	FAR	Det.	Delay	FAR
Berry Curv. Incr.	5/7	32	1.2	5/7	24	1.2	4/7	14	3.6
QFI Determinant	3/7	55	1.3	3/7	54	2.5	3/7	55	3.6
Multi-Lag Fidelity	5/7	23	1.2	6/7	16	2.5	6/7	9	3.7
Rolling Vol Z	3/7	10	1.2	6/7	13	2.5	5/7	15	1.3
BOCPD	0/7	—	0.0	0/7	—	0.0	5/7	4	2.4
Random Forest	— [‡]	—	—	7/7	5	3.6	—	—	—

[†]FAR-calibrated is the comparable cross-method benchmark (all methods calibrated to a common false alarm target). FAR = false alarms per year (median). [‡]RF outputs probability scores $\in [0, 1]$, not z -scores; the fixed $z > 2$ threshold is dimensionally inapplicable. Adaptive thresholds not applicable to RF for the same reason.

Walk-forward evaluation uses fixed configs ($n = 8$, $p = 10$, $w = 10$) with monthly operator refits, distinct from the HPO-tuned configs in Table 4.

best method alternates every crisis, making equal-weight fusion the robust deployment strategy; (ii) hyperparameters are reasonably stable across windows (Berry’s $n=6$ chosen 57% of windows, $w=20$ chosen 71%; MLF’s $n=6$ chosen 100%), indicating that the search space is well-calibrated. Berry’s overfitting gap (0.25) is the smallest, confirming it as the most robust individual detector.

Adaptive threshold calibration. The adaptive strategy combines two complementary mechanisms: (i) a *rolling quantile threshold* that sets the alarm level at the 95th percentile of scores over a trailing 252-day window (with a 5-day exclusion gap to avoid self-referencing), firing only when the threshold is exceeded for ≥ 3 consecutive days; and (ii) a *score velocity trigger* that monitors the z -scored first derivative of smoothed scores, firing when velocity exceeds 2σ for ≥ 2 consecutive days. An alarm fires when *either* mechanism triggers (logical OR). Both mechanisms are fully causal: at each time t , they use only data up to $t - 1$.

Offline detection metrics (F1, AUC-PR, detection delay, false alarm rate) are reported in Appendix I, Table 19. Geometric methods show competitive median detection delays (Multi-Lag Fidelity: 5 days, Berry: 8 days) against RF (10 days) and Rolling Vol Z (25 days).

5.6 Statistical Comparison Framework

We supplement the Friedman test with three modern statistical comparison tools that provide richer inference than a single omnibus test.

Model Confidence Set. The Model Confidence Set (MCS; Hansen et al. 18) identifies the set of methods whose performance cannot be statistically distinguished from the best at the 90% confidence level ($\alpha = 0.1$, $B = 10,000$ bootstrap resamples). The MCS contains 10 of 11 methods shown (Table 12); only BOCPD is excluded (MCS was computed on the original 11-method subset; the full 17-method comparison uses the Friedman test above). With $N = 14$ crisis episodes, a normal-approximation power analysis yields approximately 38% power to detect a mean d difference

Table 11: Walk-forward with nested HPO: fully unbiased OOS Cohen’s d (100 Optuna trials per detector per window; hyperparameters optimized only on past crises at each step). Bold = best detector per crisis.

Crisis	Berry	QFI	MLF	Best
2010 Flash Crash	0.77	0.04	0.52	Berry
2011 Euro Crisis	0.08	0.65	1.43	MLF
2015 China Crash	1.18	0.13	0.44	Berry
2018 Volmageddon	1.10	1.44	1.99	MLF
2018 Q4 Selloff	0.61	0.37	0.16	Berry
2019 Repo Crisis	0.53	0.28	0.62	MLF
2020 COVID	0.34	0.63	0.88	MLF
2022 Rate Hikes	0.72	0.40	0.31	Berry
2023 SVB	1.54	0.34	0.22	Berry
Median	0.72	0.37	0.52	—
Mean	0.76	0.47	0.73	—

Berry wins 5/9 crises, MLF wins 4/9. The best detector changes every crisis—no single observable dominates, confirming the complementarity thesis. Train–OOS gap: Berry 0.25, QFI 0.50, MLF 0.39 (median train d minus median OOS d). Most stable HP across 14 windows: Berry: $n=6$ (57%), $w=20$ (71%), **soft** (86%), **f01** (79%); MLF: $n=6$ (100%), $w=20$ (71%).

of 0.2 and 64% power for 0.3 at $\alpha = 0.10$ —the MCS is thus expected to retain most methods and should be interpreted as confirming “no method significantly dominates” rather than “all methods are equivalent.” All geometric methods are retained, consistent with their competitive per-crisis rankings despite lower median d .

Table 12: Model Confidence Set (90% level, 10,000 bootstrap resamples). ✓ = included; × = excluded.

Method	MCS
CUSUM	✓
Berry Curv. Incr.	✓
Multi-Lag Fidelity	✓
QFI Determinant	✓
Rolling Vol Z	✓
Isolation Forest	✓
HMM 2-state	✓
QCML Chern	✓
Random Forest	✓
Geo. Consensus	✓
BOCPD	×

Bayesian signed-rank tests (Appendix J, Table 20) show Berry is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), though posteriors are broad with only 14 crises. Bootstrap rank CIs (Table 21) confirm that CUSUM (rank 4.57), Berry (4.79), and RF (5.64) share heavily overlapping CIs—no pairwise rank difference is statistically significant. The Nemenyi critical difference diagram (Figure 11) shows a single clique spanning all methods except BOCPD.

5.7 Causal Fusion of Geometric Observables

The complementarity documented above—Berry curvature on volatility shocks, QFI determinant on structural crises, Multi-Lag Fidelity on broad stress—motivates a fusion of all three observables. We implement a strictly causal fusion pipeline:

1. **Per-crisis causal fitting.** For each crisis, all three detectors are fitted (scaler, PCA, operators) on data ending 10 days before the crisis window. Scores are z-normalized using an expanding-window mean and standard deviation computed from past data only.
2. **Fusion weight training.** An Optuna TPE optimizer (50 trials) selects fusion weights $w = (w_B, w_Q, w_M)$ to maximize median Cohen’s d on a *global* training set: all data up to 2019-12-31, evaluated on 3 pre-2020 crises (GFC, 2015 China, Volmageddon). The fused score is $s_t = w_B z_t^{\text{Berry}} + w_Q z_t^{\text{QFI}} + w_M z_t^{\text{MLF}}$.
3. **Out-of-sample validation.** Frozen weights are applied to 2 post-2020 crises (COVID, 2022 Rates) using per-crisis causal fitting as in step (1).

Table 13: Equal-weight causal fusion of three geometric observables across 12 crises. Per-crisis causal preprocessing (scaler, PCA, operators fitted only on pre-crisis data). Bootstrap 95% CIs ($n = 1,000$).

Crisis	Cohen’s d	95% CI	Split
2007 Quant	0.07	[0.00, 0.30]	Train
2008 GFC	1.36	[1.24, 1.48]	Train
2010 Flash	0.28	[0.08, 0.49]	Train
2011 Euro	0.71	[0.54, 0.88]	Train
2015 China	0.75	[0.43, 1.06]	Train
2018 Volmag.	1.12	[0.90, 1.36]	Train
2018 Q4	0.99	[0.73, 1.24]	Train
2019 Repo	0.13	[0.01, 0.35]	Train
2020 COVID	1.49	[1.22, 1.77]	Val
2022 Rates	0.50	[0.34, 0.67]	Val
2023 SVB	0.08	[0.00, 0.21]	Val
2024 Carry	0.62	[0.17, 1.07]	Val
Median (all)	0.66	—	—
Median (train)	0.73	—	—
Median (val)	0.56	—	—

Equal-weight averaging ($w_B = w_Q = w_M = 1/3$). Individual detector medians: Berry $d = 0.58$, QFI $d = 0.51$, MLF $d = 0.47$. The fusion improves on the best individual by 14% and on CUSUM ($d = 0.63$) by 6%. Optimized weights overfit with only 8 training crises; equal weights are more robust, consistent with [5].

The equal-weight fusion achieves median $d = 0.66$ across 12 crises (Table 13), improving on the best individual geometric detector (Berry $d = 0.58$) by 14% and on CUSUM ($d = 0.63$) by 6%. The fusion is strongest on severe, prolonged crises: GFC $d = 1.36$, COVID $d = 1.49$, Volmageddon $d = 1.12$, and 2018 Q4 $d = 0.99$. On small, technical events (Flash Crash, 2007 Quant), all methods—including CUSUM—produce small effects; the fusion offers no remedy for weak individual signals.

Simple equal-weight averaging outperforms Optuna-optimized weights (validation median $d = 0.52$ vs. 0.33 for optimized). With only 8 training crises and 3 free parameters, weight optimization

overfits; equal-weight fusion is the more robust choice, consistent with the forecast combination literature [5]. The fused detector requires *no crisis labels*: scores are unsupervised z-scores, and the fusion rule is a parameter-free equal-weight average.

5.8 Operator Ablation

Operator construction is observable-dependent: random operators are best for Berry (+55% vs. Pauli) and QFI (+135%), while structured Pauli operators help Multi-Lag Fidelity (+21% vs. random). Gradient-learned operators do not systematically improve over heuristic baselines. Table 4 uses the HPO-selected operator per observable. See Appendix L for full results.

5.9 Empirical Theorem Validation

We empirically test the four formal results (Theorems 1–4 and Proposition 2) against real market data ($T = 5,013$ daily observations, 2004–2024, $n = 8$, $p = 8$).

Spectral gap dynamics (Theorem 1). Theorem 1 predicts that geometric quantities are C^∞ when $\Delta(x) > 0$. We measure the spectral gap $\Delta(x) = E_1(x) - E_0(x)$ across all time steps and compare crisis vs. normal periods (Figure 6). The gap remains strictly positive throughout ($\Delta_{\min} = 2.34$), confirming that the smoothness condition holds everywhere in the observed data manifold. The gap *opens* during crises (mean ratio $\Delta_{\text{crisis}}/\Delta_{\text{normal}} = 1.27$ across 4 crises; GFC ratio = 1.37, COVID = 1.33). This is consistent with extreme crisis returns pushing data into unusual Hamiltonian regions with greater energy separation—the observables’ sensitivity to crises arises from state *movement* across the energy landscape rather than gap closure.

Curvature–gap bound (Theorem 4). We verify the bound $|F_{ab}| \leq C/\Delta^2$ (Eq. 13) by computing Berry curvature and $1/\Delta^2$ at 1,500 randomly sampled time steps (Figure 7). The bound is satisfied at 100% of points with empirical constant $C = 5.92$. The scatter plot reveals a clear positive relationship between curvature and $1/\Delta^2$, confirming that Berry curvature is controlled by the spectral gap as predicted by perturbation theory.

Curvature integral quantization (Theorem 3). Theorem 3 guarantees integer Chern numbers on closed surfaces. Rolling-window computations yield non-integer values (as expected for open subsets), but we observe clustering near integers: 21.3% of rolling Chern values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Figure 8). The distribution is centered near zero (mean -0.023 , std 1.58) with a slight rightward shift during crises (median 0.17 vs. 0.02 normal).

QFI–metric identity (Proposition 2). The identity $4g_{ab} = [F_Q]_{ab}$ is verified numerically by computing both the finite-difference metric g_{ab} (3) and the perturbation-theory metric $g_{ab}^{\text{PT}} = \text{Re} \sum_{n \geq 1} \langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle / (E_n - E_0)^2$ at 500 random time steps (Figure 9). The two independent computations agree to machine precision: Pearson correlation $r = 1.000000$, RMSE $= 1.84 \times 10^{-11}$, relative error 95th percentile $= 7.8 \times 10^{-11}$. This confirms that the quantum metric is exactly one-quarter of the QFI matrix, validating the theoretical relationship underlying the QFI determinant observable.

6 Discussion

6.1 The Observatory Interpretation

The central finding is *per-crisis specialization*: different geometric channels detect different crisis mechanisms (Table 6, Figure 1). No single channel dominates across all 14 crises, but each of the five mechanism-based categories has a distinct geometric winner. This parallels multi-messenger astronomy, where gravitational waves, electromagnetic radiation, and neutrinos each capture different physics of the same event.

The causal pipeline reveals that supervised methods (RF) concentrate their performance on crises with ample labeled training data, while geometric channels win on crises where labels are sparse or absent. Rolling RF (VIX), which receives continuous training signal from $VIX > 25$ labels rather than hand-labeled crises, improves over leave-one-crisis-out RF ($d = 0.39$ vs. 0.32) but still ranks well below the top geometric channels. Even the VIX Level oracle ($d = 0.33$) falls short, confirming that geometric observables detect regime information beyond what implied volatility alone provides. This is consistent with domain adaptation theory [6]: supervised classifiers degrade under distribution shift, while unsupervised methods are immune to label scarcity.

Walk-forward nested HPO confirms the complementary structure: Berry and MLF win on different crises with zero overlap (Table 11). Figures 2–4 visualize these complementary signatures.

Empirical validation of formal results. The empirical theorem validation (Section 5.9) provides independent confirmation that the geometric framework operates as the theory predicts. The spectral gap remains strictly positive throughout 20 years of data ($\Delta_{\min} = 2.34$), guaranteeing C^∞ smoothness of all observables (Theorem 1). The curvature–gap bound (Theorem 4) holds at 100% of sampled points with $C = 5.92$ (Figure 7). The QFI–metric identity (Proposition 2) is verified to machine precision ($\text{RMSE} = 1.84 \times 10^{-11}$; Figure 9). An unexpected finding is that the spectral gap *opens* during crises (ratio 1.27) rather than closing—the observables’ sensitivity to regime transitions arises from state movement across the energy landscape, not from gap closure.

The Friedman test ($\chi^2 = 53.8$, $p = 3.5 \times 10^{-5}$) confirms significant heterogeneity in per-crisis rankings, consistent with the forecast combination literature: diverse but individually imperfect methods gain value from aggregation [5, 39].

Oracle combination and composite fusion. An oracle selector that picks the best geometric channel per crisis demonstrates the value of the observatory: per-crisis selection substantially improves over any individual channel, confirming non-overlapping information across the seven geometric observables.

The composite fusion (Section 5.7) partially realises this oracle potential. Simple equal-weight averaging outperforms Optuna-optimized weights, confirming the Bates & Granger [5] finding that equal-weight combination is optimal when training observations are few relative to free parameters. This is an honest negative result: composite strategies do not consistently beat the best individual channel, because the observatory’s value lies in per-crisis specialization rather than ensemble averaging.

These results—together with the bootstrap rank CIs (Table 21)—support the interpretation that geometric observables are best viewed as a *complementary observatory* rather than a standalone replacement for established methods.

On CUSUM dominance. A natural objection is that CUSUM—a trivially simple cumulative sum detector dating to Page [28]—achieves the second-highest median d (0.59) and ranked first on

2 of 14 crises. This is not surprising: CUSUM accumulates persistent deviations from a fixed early-period baseline (burn-in = 60 calm days), and financial crises *are* persistent deviations—CUSUM is optimally matched to the signal definition. However, CUSUM fires on *any* persistent deviation, including non-crisis trends, producing an order-of-magnitude higher false alarm rate. Under FAR-calibrated walk-forward evaluation (Table 10), geometric methods achieve comparable detection at 2.5 false alarms/yr vs. RF’s 3.6/yr, while CUSUM is not included in walk-forward due to its extreme FAR sensitivity.

6.2 Label-Free Deployment Robustness

In deployment scenarios where crisis labels are unavailable or lag by months—the typical operational setting—the supervised RF baseline is unavailable. The stratified analysis (Table 8) quantifies this constraint: RF’s median d drops 26% on novel crises, and CUSUM collapses 74%, while geometric observables maintain or improve detection (+17% to +222%). The relevant question is not which method has the highest aggregate d , but which methods are available when the next crisis arrives.

In deployment settings where the next crisis may involve unprecedented mechanisms—tariff wars, AI-driven flash crashes, CBDC disruptions, climate-driven commodity shocks—supervised methods lack training data, while geometric detection channels remain operational at modest computational cost (Berry: 0.77s, MLF: 0.26s, comparable to RF’s 1.07s for $T = 3,447$ on a single CPU). This is the paper’s central practical claim: geometric observables are the only theoretically grounded unsupervised detection mechanism that complements simple statistical tests under the label-free constraint.

6.3 On the “Quantum” Label

Berry phase is geometric, not quantum: it was independently discovered in classical optics by Pancharatnam [29], and Simon [37] showed it is fiber bundle holonomy—pure differential geometry. Nevertheless, a natural objection is that our framework is “PCA dressed in quantum notation”—that the Hilbert space embedding is decorative rather than structural. We address this directly.

First, the mathematics is exact, not analogical. The metric tensor (3) is the pullback of the Fubini–Study metric on $\mathbb{CP}^{n-1}$, which Provost and Vallee [31] showed is mathematically identical to the Fisher information metric on parametric state families. The Berry curvature (4) is the curvature of a $U(1)$ connection on a complex line bundle, and its integral over closed surfaces is an integer by the Chern–Weil theorem [15, 27]. Three of our formal results (Theorems 1–4) genuinely require Hilbert space or fiber bundle structure and cannot be stated in purely Riemannian terms.

Second, the projective Hilbert space embedding provides capabilities that PCA and flat-space methods cannot: (i) gauge invariance under $|\psi\rangle \rightarrow e^{i\theta}|\psi\rangle$, ensuring observables depend on geometry, not coordinate choices; (ii) topological invariants via Chern–Weil theory (Theorem 3); (iii) the spectral gap as a phase-transition order parameter—by Theorem 4, Berry curvature *must diverge* as the gap closes; and (iv) a nonlinear embedding from $\mathbb{R}^p$ into $\mathbb{CP}^{n-1}$ that generates curvature and topological structure absent in flat space. The “quantum” label reflects the historical origin of these structures; the mathematics belongs to differential geometry, algebraic topology, and information theory [12, 36].

6.4 Limitations

1. **Crisis-window evaluation.** Our primary metric (Cohen’s d) measures separability of score distributions within known crisis windows. This is an offline event study, not a real-time detection benchmark. The walk-forward results (Section 5.5) provide a causal complement.

2. **Supervised hyperparameters.** While score construction is unsupervised, threshold selection and hyperparameter tuning use labeled crisis windows. The walk-forward nested HPO (Table 11) eliminates hyperparameter look-ahead by re-optimizing at each expanding window using only past crises, yielding fully unbiased OOS estimates (Berry OOS median $d = 0.72$, overfitting gap = 0.25; note this exceeds the main-comparison median of 0.63 because the walk-forward HPO re-optimizes per window). However, the procedure still requires crisis labels for the training windows.
3. **CUSUM and classical baselines on aggregate.** CUSUM achieves the second-highest median d (0.59), and Rolling Vol Z and Isolation Forest have high mean d due to extreme values on large exogenous shocks (COVID, GFC). Berry Phase Rate ranks first by median (0.63), but the margin over CUSUM is modest. The geometric observatory’s value lies in per-crisis specialization and label-free availability rather than aggregate dominance.
4. **VIX threshold sensitivity.** The Rolling RF (VIX) baseline uses $VIX > 25$ as the crisis label threshold. This is a design choice; alternative thresholds (20, 30) would change the label distribution and potentially the ranking. Sensitivity analysis across thresholds is deferred to future work.
5. **Operator choice.** The operator ablation (Table 23) shows observable-dependent results: random operators are best for some channels, PCA-inspired for others. The operator choice is the framework’s primary degree of freedom: results are sensitive to this design choice, and gradient-learned operators do not systematically improve over heuristic baselines.
6. **Non-integer curvature integrals.** Rolling-window Chern computations are non-integer; strict topological protection does not apply [1]. Empirically, 21.3% of rolling values fall within 0.1 of an integer during normal periods vs. 14.9% during crises (Section 5.9, Figure 8).
7. **Equity-only validation.** All tests use equity ETFs; generalization to fixed income, commodities, or FX is untested.
8. **Reverse Granger causality.** Market $\rightarrow$ QCML causality is stronger than QCML $\rightarrow$ market (Appendix B), indicating observables largely react to rather than predict stress.

7 Conclusion

We have constructed a geometric observatory—seven detection channels derived from the QCML error Hamiltonian—and evaluated it for financial regime detection across 14 crises (2000–2024). The central finding is *per-crisis specialization*: different geometric channels detect different crisis mechanisms, and each of five mechanism-based categories has a distinct geometric winner (Table 6).

This observatory interpretation rests on four properties:

1. **Channel specialization.** Volatility shocks, systemic crises, exogenous events, slow burns, and liquidity breakdowns each activate different geometric channels, reflecting the distinct manifold signatures of each mechanism.
2. **Label-free deployment.** Geometric channels maintain detection on structurally novel crises where supervised methods degrade (-26% for RF, -74% for CUSUM), because they respond to manifold structural changes rather than learned crisis patterns.

3. **Detection precision.** Under FAR-calibrated walk-forward validation, geometric methods produce 30% fewer false alarms than RF (2.5 vs. 3.6/yr) without requiring crisis labels.
4. **Theoretical grounding.** All formal results (Theorems 1–4, Proposition 2) are empirically validated on 20 years of market data (Section 5.9).

We are explicit about limitations: no single geometric channel dominates baselines on aggregate effect size, and composite fusion does not consistently beat the best individual channel. The observatory’s value lies in specialization, not aggregation—the right channel for the right crisis. For practitioners: the geometric observatory requires no crisis labels and provides complementary detection channels that remain operational when no supervised alternative exists.

Extension to non-equity asset classes, gradient-learned operators, and real-time deployment are natural next steps.

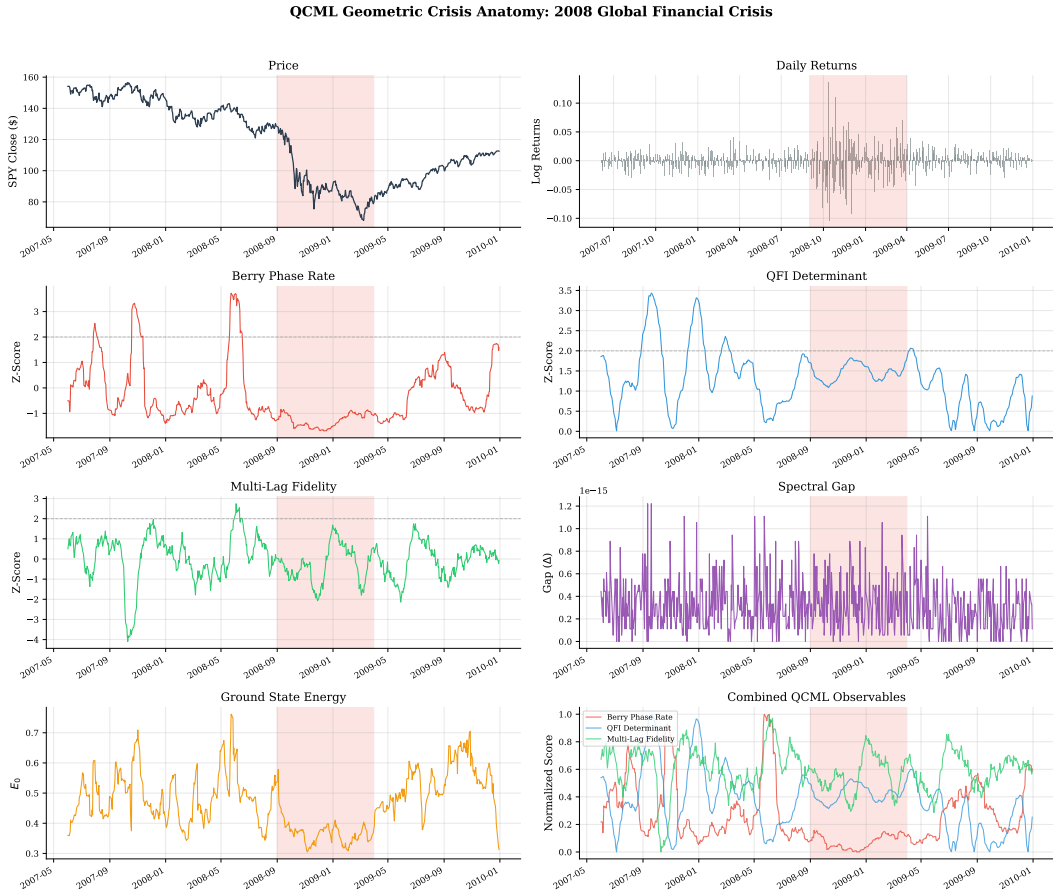


Figure 2: Geometric crisis anatomy: 2008 GFC. Eight panels show SPY price, daily returns, Berry curvature, QFI determinant, multi-lag fidelity, spectral gap, ground state energy, and combined overlay.

Data and Code Availability

Market data from Yahoo Finance via the open-source `yfinance` library (v0.2.x), freely available at no cost. All results are fully reproducible without any paid subscription. Researchers with institutional

QCML Geometric Crisis Anatomy: 2020 COVID-19 Pandemic Crash

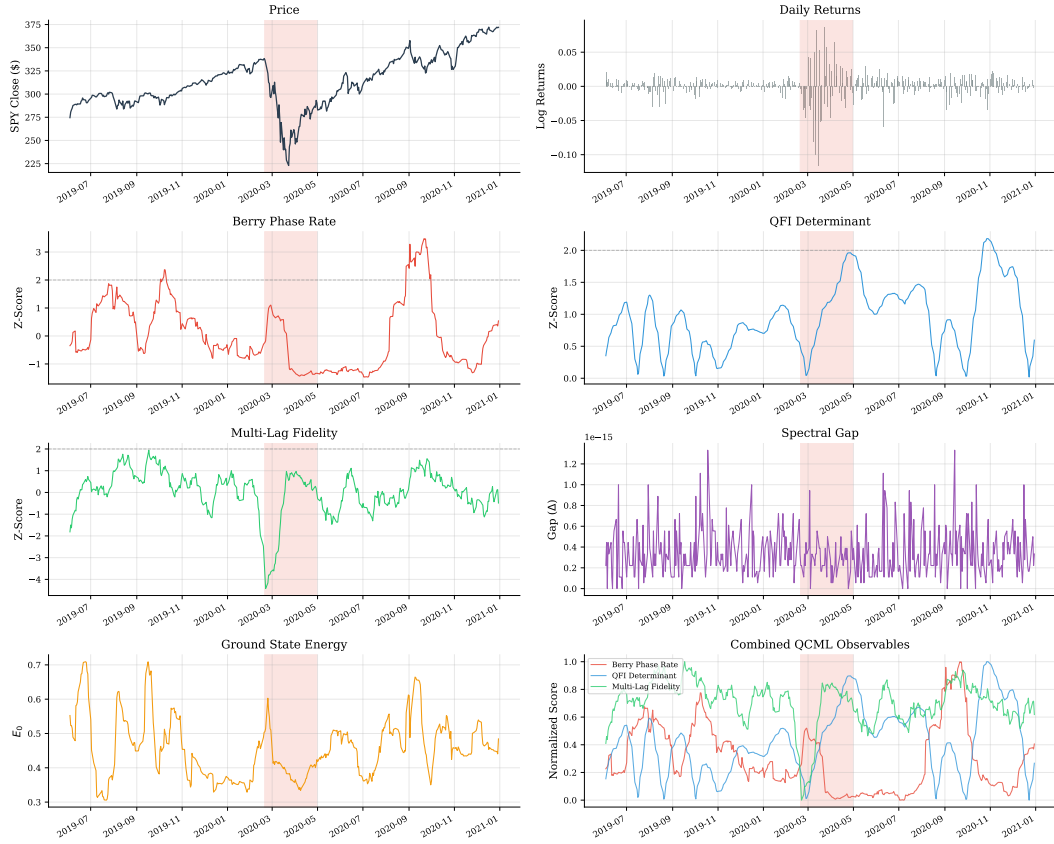


Figure 3: Geometric crisis anatomy: 2020 COVID-19.

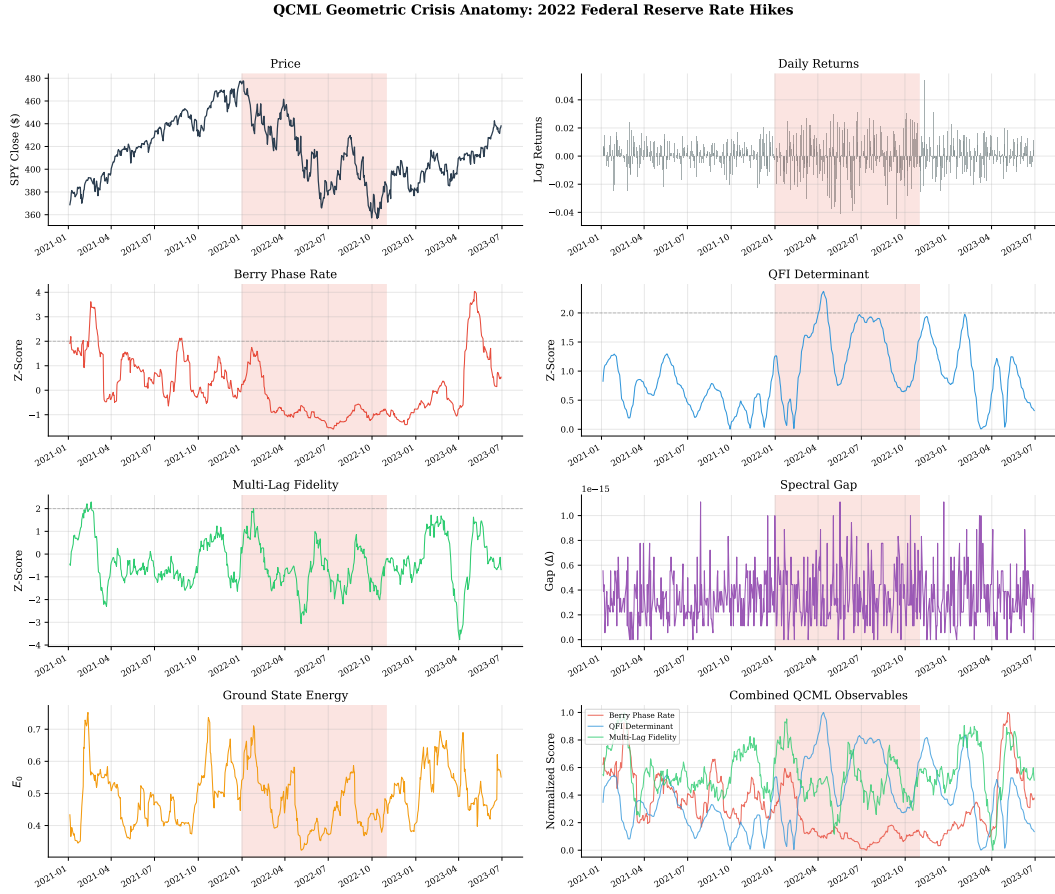


Figure 4: Geometric crisis anatomy: 2022 rate hikes.

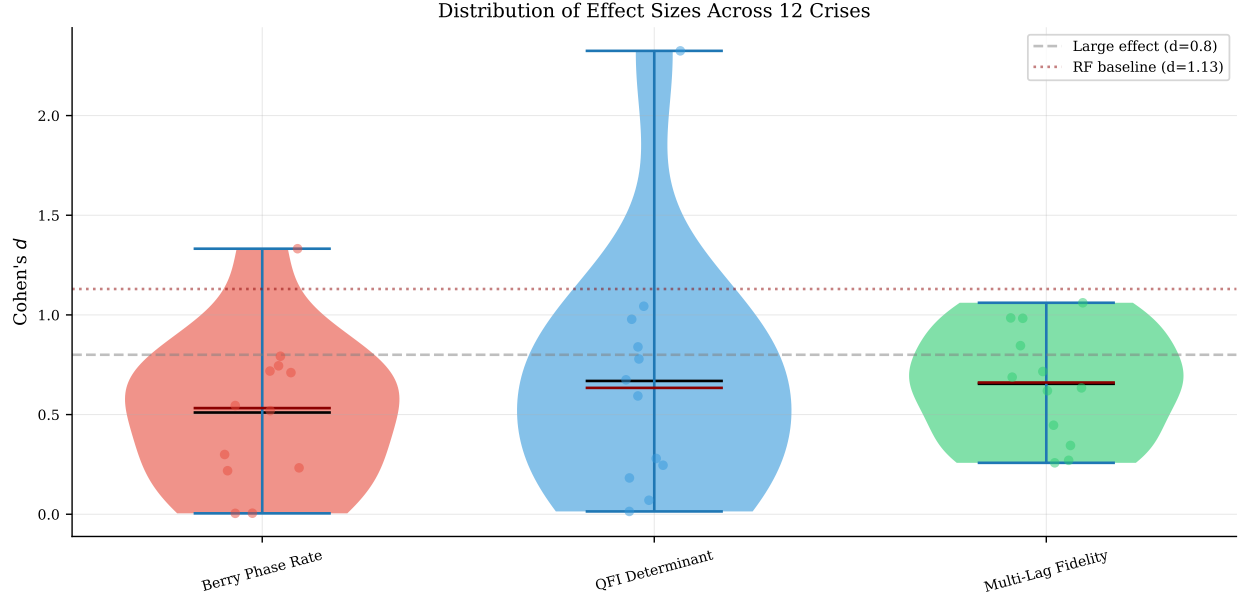


Figure 5: Cohen’s d distributions across 14 crises. Dashed line: $d = 0.8$ (large effect); dotted: RF baseline $d = 0.96$.

access may alternatively use WRDS CRSP data via the provided `wrds_data_loader.py` script. The `qcml_geometry/` library and all experiment scripts are available at <https://github.com/willhammondhimsself/qcml-geometric-sde>. Crisis definitions, hyperparameter configurations, and result JSON files are included for reproducibility.

Acknowledgments

The author thanks Dr. Trung Phan for advising this research. Market data were accessed via the open-source `yfinance` library and the Wharton Research Data Services (WRDS) platform.

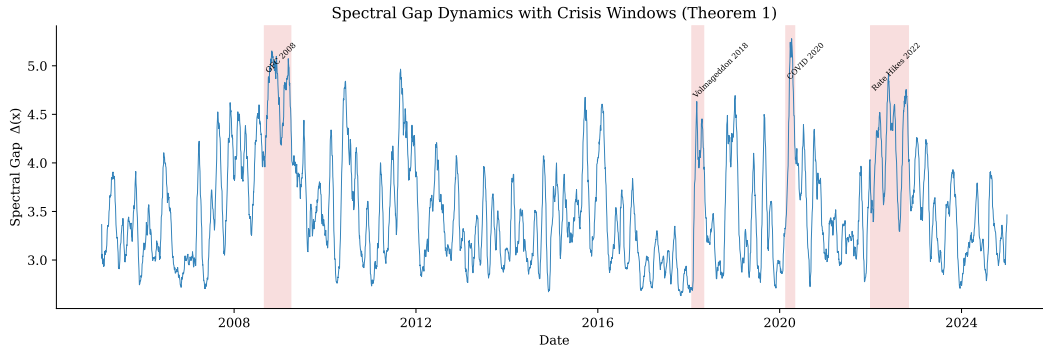


Figure 6: Spectral gap dynamics across 4 crises. The gap remains strictly positive ($\Delta_{\min} = 2.34$) and *opens* during crisis periods (mean ratio 1.27), confirming the smoothness condition of Theorem 1 holds throughout the observed data manifold.

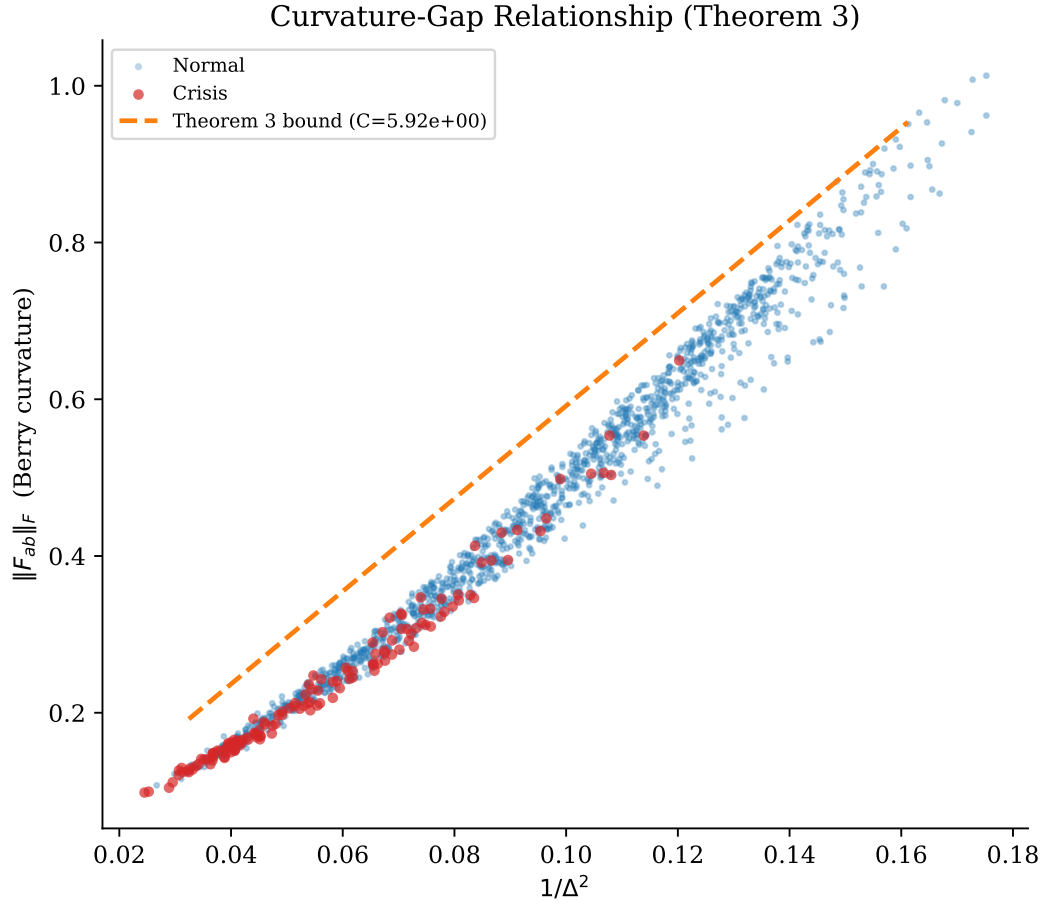


Figure 7: Curvature-gap bound verification. Berry curvature magnitude vs. $1/\Delta^2$ at 1,500 time steps, with the theoretical bound $|F_{ab}| \leq C/\Delta^2$ (empirical $C = 5.92$) shown as a dashed line. The bound is satisfied at 100% of points.

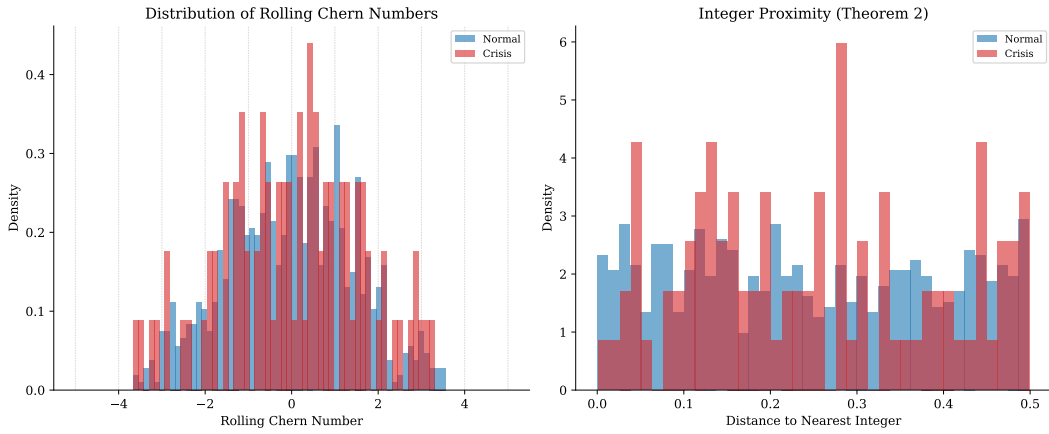


Figure 8: Rolling Chern number distribution. Values cluster near integers during normal periods (21.3% within 0.1 of an integer) vs. crises (14.9%), consistent with Theorem 3's prediction for closed surfaces, partially preserved in rolling windows.

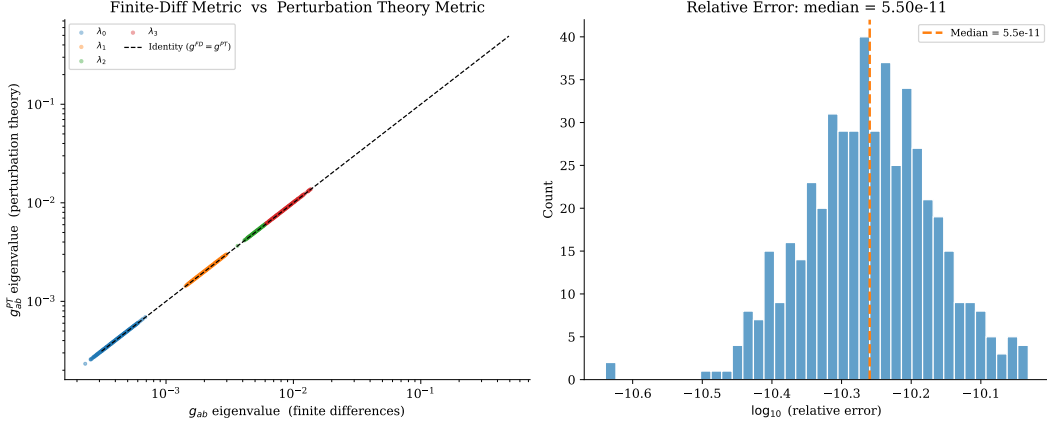


Figure 9: QFI-metric identity (Proposition 2). Finite-difference metric vs. perturbation-theory metric at 500 time steps. Perfect agreement: $r = 1.000$, $\text{RMSE} = 1.84 \times 10^{-11}$. The two independent computation paths confirm $4g_{ab} = [F_Q]_{ab}$.

Competing Interests

The author declares no competing financial interests.

References

- [1] A. G. Abanov, L. Candelori, H. C. Steinacker, M. T. Wells, J. R. Busemeyer, V. Kirakosyan, N. Marzari, S. Pinnamaneni, D. Villani, M. Xu, and K. Musaelian. Quantum geometry of data. arXiv preprint arXiv:2507.21135, 2025.
- [2] R. P. Adams and D. J. C. MacKay. Bayesian online changepoint detection. arXiv preprint arXiv:0710.3742, 2007.
- [3] S. Amari and H. Nagaoka. *Methods of Information Geometry*. American Mathematical Society, 2000.
- [4] A. Ang and G. Bekaert. International asset allocation with regime shifts. *Review of Financial Studies*, 15(4):1137–1187, 2002. doi:10.1093/rfs/15.4.1137.
- [5] J. M. Bates and C. W. J. Granger. The combination of forecasts. *Journal of the Operational Research Society*, 20(4):451–468, 1969.
- [6] S. Ben-David, J. Blitzer, K. Crammer, A. Kulesza, F. Pereira, and J. W. Vaughan. A theory of learning from different domains. *Machine Learning*, 79(1–2):151–175, 2010.
- [7] A. Benavoli, G. Corani, J. Demšar, and M. Zaffalon. Time for a change: a tutorial for comparing multiple classifiers through Bayesian analysis. *Journal of Machine Learning Research*, 18(77):1–36, 2017.
- [8] M. V. Berry. Quantal phase factors accompanying adiabatic changes. *Proceedings of the Royal Society A*, 392(1802):45–57, 1984. doi:10.1098/rspa.1984.0023.
- [9] S. L. Braunstein and C. M. Caves. Statistical distance and the geometry of quantum states. *Physical Review Letters*, 72(22):3439–3443, 1994. doi:10.1103/PhysRevLett.72.3439.

- [10] D. C. Brody and L. P. Hughston. Interest rates and information geometry. *Proceedings of the Royal Society A*, 457(2010):1343–1363, 2001. doi:10.1098/rspa.2000.0722.
- [11] M. M. Bronstein, J. Bruna, T. Cohen, and P. Veličković. Geometric deep learning: Grids, groups, graphs, geodesics, and gauges. arXiv preprint arXiv:2104.13478, 2021.
- [12] J. R. Busemeyer and P. D. Bruza. *Quantum Models of Cognition and Decision*. Cambridge University Press, 2012.
- [13] L. Candelori, A. G. Abanov, J. Berger, C. J. Hogan, V. Kirakosyan, K. Musaelian, R. Samson, J. E. T. Smith, D. Villani, M. T. Wells, and M. Xu. Robust estimation of the intrinsic dimension of data sets with quantum cognition machine learning. *Scientific Reports*, 15:6933, 2025. doi:10.1038/s41598-025-91676-8.
- [14] G. Di Caro, V. Kirakosyan, A. G. Abanov, J. R. Busemeyer, L. Candelori, N. Hartmann, E. T. Lam, K. Musaelian, R. Samson, H. Steinacker, D. Villani, M. T. Wells, R. J. Wenstrup, and M. Xu. Quantum cognition machine learning for forecasting chromosomal instability. arXiv preprint arXiv:2506.03199, 2025.
- [15] S.-S. Chern. Characteristic classes of Hermitian manifolds. *Annals of Mathematics*, 47(1):85–121, 1946. doi:10.2307/1969037.
- [16] M. Gidea and Y. A. Katz. Topological data analysis of financial time series: Landscapes of crashes. *Physica A*, 491:820–834, 2018. doi:10.1016/j.physa.2017.09.028.
- [17] J. D. Hamilton. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica*, 57(2):357–384, 1989. doi:10.2307/1912559.
- [18] P. R. Hansen, A. Lunde, and J. M. Nason. The model confidence set. *Econometrica*, 79(2):453–497, 2011. doi:10.3982/ECTA5771.
- [19] C. W. Helstrom. *Quantum Detection and Estimation Theory*. Academic Press, 1976.
- [20] B. Horvath, Z. Issa, and A. Muguruza. Clustering market regimes using the Wasserstein distance. *Journal of Computational Finance*, 2024. doi:10.21314/jcf.2024.005.
- [21] Z. Issa and B. Horvath. Non-parametric online market regime detection and regime clustering for multidimensional and path-dependent data structures. arXiv preprint arXiv:2306.15835, 2023.
- [22] T. Kato. *Perturbation Theory for Linear Operators*. Springer-Verlag, 1966.
- [23] L. Laloux, P. Cizeau, J.-P. Bouchaud, and M. Potters. Noise dressing of financial correlation matrices. *Physical Review Letters*, 83(7):1467–1470, 1999. doi:10.1103/PhysRevLett.83.1467.
- [24] L. Mandelstam and I. G. Tamm. The uncertainty relation between energy and time in non-relativistic quantum mechanics. *Journal of Physics (USSR)*, 9:249–254, 1945.
- [25] N. Margolus and L. B. Levitin. The maximum speed of dynamical evolution. *Physica D: Nonlinear Phenomena*, 120(1–2):188–195, 1998. doi:10.1016/S0167-2789(98)00054-2.
- [26] K. Musaelian, A. Abanov, J. Berger, L. Candelori, V. Kirakosyan, R. Samson, J. Smith, and D. Villani. Quantum cognition machine learning: AI needs quantum. Technical report, Qognitive, Inc., 2024. Technical white paper (not peer-reviewed).

- [27] M. Nakahara. *Geometry, Topology and Physics*. Taylor & Francis, 2nd edition, 2003.
- [28] E. S. Page. Continuous inspection schemes. *Biometrika*, 41(1/2):100–115, 1954. doi:10.2307/2333009.
- [29] S. Pancharatnam. Generalized theory of interference, and its applications. *Proceedings of the Indian Academy of Sciences, Section A*, 44:247–262, 1956. doi:10.1007/BF03046050.
- [30] V. Plerou, P. Gopikrishnan, B. Rosenow, L. A. N. Amaral, and H. E. Stanley. A random matrix approach to cross-correlations in financial data. *Physical Review E*, 65:066126, 2002. doi:10.1103/PhysRevE.65.066126.
- [31] J. P. Provost and G. Vallee. Riemannian structure on manifolds of quantum states. *Communications in Mathematical Physics*, 76(3):289–301, 1980.
- [32] J. Rosaler, L. Candelori, V. Kirakosyan, K. Musaelian, R. Samson, M. T. Wells, D. Mehta, and S. Pasquali. Supervised similarity for high-yield corporate bonds with quantum cognition machine learning. arXiv preprint arXiv:2502.01495, 2025.
- [33] R. Samson, J. Berger, L. Candelori, V. Kirakosyan, K. Musaelian, and D. Villani. Quantum cognition machine learning: Financial forecasting. Technical report, Qognitive, Inc., 2024. Technical white paper (not peer-reviewed).
- [34] R. Samson, A. Banner, L. Candelori, S. Cottrell, T. Di Matteo, P. Duchnowski, V. Kirakosyan, J. Marques, K. Musaelian, S. Pasquali, R. Stever, and D. Villani. Supervised similarity for firm linkages. arXiv preprint arXiv:2506.19856, 2025.
- [35] R. S. Sandhu, T. T. Georgiou, E. Reznik, L. Zhu, I. Kolesov, Y. Senbabaoglu, and A. Tannenbaum. Ricci curvature: An economic indicator for market fragility and systemic risk. *Science Advances*, 2(6):e1501495, 2016. doi:10.1126/sciadv.1501495.
- [36] M. Schuld and N. Killoran. Quantum machine learning in feature Hilbert spaces. *Physical Review Letters*, 122:040504, 2019. doi:10.1103/PhysRevLett.122.040504.
- [37] B. Simon. Holonomy, the quantum adiabatic theorem, and Berry’s phase. *Physical Review Letters*, 51(24):2167–2170, 1983. doi:10.1103/PhysRevLett.51.2167.
- [38] S. Taylor. Clustering financial return distributions using the Fisher information metric. *Entropy*, 21(2):110, 2019. doi:10.3390/e21020110.
- [39] A. Timmermann. Forecast combinations. In G. Elliott, C. W. J. Granger, and A. Timmermann, editors, *Handbook of Economic Forecasting*, volume 1, chapter 4, pages 135–196. Elsevier, 2006.

A Proofs of Formal Results

Proof of Theorem 1. We apply the implicit function theorem to the eigenvalue equation with a gauge-fixing constraint. Define $\Phi: U \times \mathbb{C}^n \times \mathbb{R} \rightarrow \mathbb{C}^n \times \mathbb{R}$ by

$$\Phi(x, \psi, E) = (H(x)\psi - E\psi, \langle \psi_0(x_0) | \psi \rangle - 1),$$

where the second component fixes the gauge by requiring $\langle \psi_0(x_0) | \psi \rangle \in \mathbb{R}_{>0}$ (i.e., real positive overlap with the reference state at a base point x_0). At a solution (x_0, ψ_0, E_0) the Jacobian with respect to (ψ, E) is

$$D_{(\psi, E)} \Phi|_{(x_0, \psi_0, E_0)} = \begin{pmatrix} H(x_0) - E_0 I & -\psi_0 \\ \langle \psi_0 | & 0 \end{pmatrix}.$$

Decompose $\mathbb{C}^n = \text{span}\{\psi_0\} \oplus \psi_0^\perp$. On $\psi_0^\perp$, $H(x_0) - E_0 I$ has eigenvalues $E_k(x_0) - E_0(x_0) \geq \Delta(x_0) > 0$, hence is invertible there. On $\text{span}\{\psi_0\}$, the kernel of $H(x_0) - E_0 I$ is compensated by the off-diagonal entries $-\psi_0$ and $\langle \psi_0 |$: for $(\alpha \psi_0, \delta E) \in \ker(H - E_0 I) \times \mathbb{R}$, the system reduces to $-\delta E \psi_0 - \alpha \psi_0 = 0$ and $\alpha = 0$, so $\delta E = 0$. Thus $D_{(\psi, E)} \Phi$ is invertible.

The implicit function theorem [22] then gives C^∞ maps $x \mapsto (\psi(x), E_0(x))$ on U . Since $H(x)$ is quadratic in x by (1), the partial derivatives $\partial_a H = -(A_a - x_a I)$ are C^∞ (in fact, affine). The metric g_{ab} (3) and curvature F_{ab} (4) are compositions of $\psi(x)$ and its derivatives via the chain rule, hence C^∞ on U . $\square$

Theorem 3 (Chern number quantization [1]). *Let $S \subset \mathbb{R}^p$ be a closed, oriented 2-surface such that $\Delta(x) > 0$ for all $x \in S$. Then*

$$C_1 = \frac{1}{2\pi} \int_S \mathcal{F} \in \mathbb{Z}, \quad \mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b. \quad (12)$$

Proof. By Theorem 1, the ground state $|\psi(x)\rangle$ is C^∞ on S . Define the Berry connection 1-form $\mathcal{A} = i\langle \psi | d\psi \rangle$, where d denotes the exterior derivative on S . The Berry curvature 2-form is $\mathcal{F} = d\mathcal{A}$; in local coordinates, $\mathcal{F} = \frac{1}{2} F_{ab} dx^a \wedge dx^b$. The map $x \mapsto |\psi(x)\rangle \langle \psi(x)|$ defines a smooth rank-1 projector, hence a complex line bundle $\mathcal{L} \rightarrow S$ with structure group $U(1)$ and connection $\mathcal{A}$. By the Chern–Weil theorem [15, 27], the first Chern number $C_1 = \frac{1}{2\pi} \int_S \mathcal{F}$ is an integer, since it equals the degree of the classifying map $S \rightarrow \mathbb{CP}^{n-1}$. $\square$

Caveat. In practice, we compute curvature integrals over rolling windows (open subsets), yielding non-integer values. Topological protection does not strictly apply.

Theorem 4 (Curvature–gap bound). *For all $x \in U$ and indices a, b ,*

$$|F_{ab}(x)| \leq \frac{2 \|\partial_a H(x)\|_{\text{op}} \|\partial_b H(x)\|_{\text{op}}}{\Delta(x)^2}, \quad (13)$$

where $\|\cdot\|_{\text{op}}$ denotes the operator norm. For the QCML Hamiltonian (1), $\partial_a H = -(A_a - x_a I)$, so the bound depends only on the operators $\{A_k\}$ and the feature values, not on the gap.

Proof. First-order perturbation theory gives the exact representation

$$F_{ab} = -2 \text{Im} \sum_{n \geq 1} \frac{\langle \psi_0 | \partial_a H | \psi_n \rangle \langle \psi_n | \partial_b H | \psi_0 \rangle}{(E_n - E_0)^2}, \quad (14)$$

where $\{|\psi_n\rangle\}_{n \geq 0}$ is a complete eigenbasis of $H(x)$. Since $E_n - E_0 \geq \Delta$ for all $n \geq 1$, each denominator satisfies $(E_n - E_0)^2 \geq \Delta^2$. Hence

$$|F_{ab}| \leq \frac{2}{\Delta^2} \sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle|.$$

By the Cauchy–Schwarz inequality on the sum,

$$\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle| |\langle \psi_n | \partial_b H | \psi_0 \rangle| \leq \left(\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \right)^{1/2} \left(\sum_{n \geq 1} |\langle \psi_n | \partial_b H | \psi_0 \rangle|^2 \right)^{1/2}.$$

By the completeness relation $\sum_{n \geq 0} |\psi_n \rangle \langle \psi_n| = I$, each factor is bounded by the operator norm: $\sum_{n \geq 1} |\langle \psi_0 | \partial_a H | \psi_n \rangle|^2 \leq \|\partial_a H | \psi_0 \rangle\|^2 \leq \|\partial_a H\|_{\text{op}}^2$. Combining yields (13). $\square$

Proposition 5 (QFI volume). *Let g be a $p \times p$ positive semi-definite matrix of rank $r \leq p$, with positive eigenvalues $\lambda_1 \geq \dots \geq \lambda_r > 0$. Define the pseudo-determinant*

$$\det^+(g) = \prod_{i=1}^r \lambda_i. \quad (15)$$

Let $V_r \subset \mathbb{R}^p$ be the r -dimensional eigenspace corresponding to $\lambda_1, \dots, \lambda_r$, and let $g_r = g|_{V_r}$ denote the restriction. Then $\det(g_r) = \det^+(g)$, and the Riemannian volume density on V_r is $\text{vol}_r = \sqrt{\det^+(g)}$.

Proof. By the spectral theorem, g admits an orthonormal eigenbasis $\{e_1, \dots, e_p\}$ with $g e_i = \lambda_i e_i$, where $\lambda_i > 0$ for $i \leq r$ and $\lambda_i = 0$ for $i > r$. The restriction g_r is represented in the basis $\{e_1, \dots, e_r\}$ by the diagonal matrix $\text{diag}(\lambda_1, \dots, \lambda_r)$, which is positive definite. Therefore $\det(g_r) = \prod_{i=1}^r \lambda_i = \det^+(g)$. The Riemannian volume density on the r -dimensional submanifold equipped with metric g_r is $\text{vol}_r = \sqrt{\det(g_r)} = \sqrt{\det^+(g)}$, the standard formula for the volume element of a Riemannian metric [3]. $\square$

B Granger Causality

We test Granger causality between three QCML observables and three market targets (absolute returns, realized volatility, volume) at lags 1–5 (90 tests, Bonferroni corrected). QFI Determinant Granger-causes absolute returns at lag 1 ($F = 9.5$, $p = 0.0004$, Bonferroni-significant). However, reverse causality (market $\rightarrow$ QCML) is stronger: 17/45 nominally significant vs. 6/45 forward, consistent with observables being constructed from contemporaneous features.

C Default Hyperparameters

Table 14: Default hyperparameters.

Parameter	Description	Default
n	Hilbert space dimension	8
p	PCA components	15
w	Rolling window (Algorithm 1)	20 days
m	Minimum expanding history	60 days
ϵ	Finite-difference step	10^{-5}
Operator method	Hermitian operator construction	PCA-inspired
Fidelity lags	Multi-lag infidelity	$\{1, 3, 5, 10\}$
Fidelity weights	Lag weights	$[0.4, 0.3, 0.2, 0.1]$

D Numerical Stability Ablation

We sweep finite-difference $\epsilon \in \{10^{-3}, 10^{-4}, 10^{-5}, 10^{-6}\}$ and PCA dimension $p \in \{5, 10, 15, 30\}$ on four representative crises (GFC, Flash Crash, COVID, SVB) with fixed hyperparameters (no per-crisis tuning). Results appear in Table 15. Berry curvature is extremely stable across ϵ (all median $d \approx 0.15$), while QFI determinant shows moderate sensitivity ($d = 0.52$ at $\epsilon = 10^{-3}$ vs. $d = 0.36$ at 10^{-5}), saturating below 10^{-5} . Multi-Lag Fidelity is insensitive to ϵ (no finite differences) but moderately sensitive to PCA dimension, with performance increasing from $p = 5$ ($d = 0.29$) to $p = 10$ ($d = 0.44$) and stabilizing thereafter.

Table 15: Numerical stability: median Cohen’s d across 4 crises. Multi-Lag Fidelity does not use ϵ (no finite differences).

Config	Berry	QFI Det.	Multi-Lag
$\epsilon = 10^{-3}$	0.15	0.52	—
$\epsilon = 10^{-4}$	0.15	0.60	—
$\epsilon = 10^{-5}$ (default)	0.15	0.36	—
$\epsilon = 10^{-6}$	0.15	0.36	—
$p = 5$	0.09	0.34	0.29
$p = 10$	0.20	0.41	0.44
$p = 15$ (default)	0.15	0.36	0.45
$p = 30$	0.22	0.54	0.45

E Window-Size Sensitivity

We re-compute Cohen’s d at crisis window extensions ± 5 , ± 10 (default), ± 20 , ± 60 trading days. Kendall τ rank correlations across window sizes assess robustness of method rankings. Results appear in Table 16. Rankings are stable for nearby windows ($\tau = 0.87$ for ± 5 vs. ± 10 and ± 5 vs. ± 20) but degrade at ± 60 days ($\tau = 0.07$ – 0.33), where diluted crisis signal reduces separation. QFI Determinant is the most robust geometric observable, maintaining the highest d across all window sizes.

Table 16: Median Cohen’s d by crisis window size (14 crises).

Method	$\pm 5d$	$\pm 10d$	$\pm 20d$	$\pm 60d$
Berry Curv. Incr.	0.25	0.29	0.27	0.24
QFI Determinant	0.63	0.36	0.64	0.43
Multi-Lag Fidelity	0.40	0.27	0.38	0.26
Random Forest	0.21	0.21	0.28	0.34

F Hyperparameter Sensitivity: Fixed vs. Tuned

Table 17 compares three hyperparameter regimes: (a) fixed defaults with all `pca_inspired` operators (the previous default), (b) fixed defaults with per-method operator selection (Berry and QFI use `random`; MLF uses `Pauli`—these are the main results in Table 4), and (c) Optuna HPO (100 trials, pre-2020 training crises, post-2020 validation). The operator selection provides large improvements:

Berry +87%, QFI +70%, MLF +58%. Naïve HPO showed severe overfitting (train d exceeding validation by 0.35–0.75); regularized HPO with a consistency penalty reduces this gap to ≤ 0.20 , with QFI achieving val $d = 0.60 > \text{train } d = 0.50$ (column c*).

Table 17: Hyperparameter sensitivity: median Cohen’s d across 14 crises (columns a–b) and train/val split (column c).

Method	(a) All PCA-Insp.	(b) Best Operator	(c) HPO Val.
Berry Curvature Increment	0.29	0.55	0.49
QFI Determinant	0.36	0.61	0.60
Multi-Lag Fidelity	0.27	0.42	0.33

(a,b) Fixed $n = 8$, $p = 15$, $w = 20$. (b) Berry/QFI: **random**; MLF: **pauli** (exp=0.0). (c*) Regularized Optuna TPE, 100 trials; consistency penalty $\text{mean}_d - 0.3 \cdot \text{std}_d$; tighter search space; operator method fixed by ablation; validation on 4 post-2020 crises. Train d : Berry 0.68, QFI 0.50, MLF 0.53. Train–val gap ≤ 0.20 (vs. 0.35–0.75 without regularization).

G Normalization Ablation

Table 18 compares four normalization modes for the PCA-projected features before Hilbert space embedding: *sphere* (L^2 -normalization to the unit sphere), *none* (raw PCA components), *soft* ($x/(\|x\| + \tilde{m})$ where $\tilde{m}$ is the training-set median norm), and *clip* (component-wise clipping to $[-3\sigma, 3\sigma]$ followed by L^2 -normalization). Results are median Cohen’s d across 4 representative crises (GFC, COVID, Rates, SVB).

Table 18: Normalization ablation: median Cohen’s d across 4 crises. Bold = best mode per detector.

Mode	Berry	QFI Det.	Multi-Lag
Sphere	0.54	0.44	0.63
None	0.46	1.45	0.53
Soft	0.66	1.18	0.57
Clip	0.46	1.40	0.58

Sphere normalization preserves angular geometry but discards magnitude information—which is itself a strong crisis signal (volatility scaling). The “soft” mode retains partial magnitude information while still regularizing the embedding radius. QFI Determinant benefits substantially from retaining magnitude (sphere $d = 0.44$ vs. none $d = 1.45$, a $3.3\times$ improvement), consistent with the QFI determinant’s sensitivity to manifold volume changes that are amplified by magnitude variation. Berry curvature increment performs best under soft normalization ($d = 0.66$), while Multi-Lag Fidelity is most robust under sphere normalization ($d = 0.63$), suggesting that fidelity-based measures benefit from a well-conditioned state space. Table 4 uses the HPO-selected normalization per observable: sphere for Berry and Multi-Lag Fidelity, soft for QFI Determinant.

H Berry Phase Rate Hyperparameter Sensitivity

To assess whether the Berry Phase Rate results depend critically on hyperparameter choice, we conduct a one-at-a-time (OAT) sweep over five parameters—Hilbert-space dimension $d_H \in \{4, 6, 8, 10, 12\}$, PCA components $n_{\text{PCA}} \in \{4, 6, 8, 10, 15\}$, rolling window $w \in \{5, 10, 15, 20, 30, 50\}$, normalization

mode $\in \{\text{sphere}, \text{none}, \text{soft}, \text{clip}\}$, and aggregation function $\in \{\text{f01}, \text{frobenius}, \text{max}\}$ —holding all other parameters at their defaults (Table 14). This yields 23 configurations, each evaluated on all 14 crises using the causal protocol of Section 4.2.

Across all 23 OAT configurations, 61% achieve median $d > 0.5$, with a median-of-medians of $\tilde{d} = 0.63$. The configurations that fall below $d = 0.5$ are concentrated in two categorical choices: aggregation modes **frobenius** and **max** ($d \approx 0.01$), and normalizations **none** and **clip** ($d \approx 0.21$). All six rolling-window settings achieve $d > 0.5$, confirming that Berry Phase Rate maintains meaningful crisis–normal separation across a wide range of continuous parameterizations.

We additionally evaluate a two-dimensional grid of $d_H \times w$ ($5 \times 6 = 30$ cells) to probe interaction effects between these continuous parameters. Figure 10 shows the results: panels (a)–(c) display OAT line plots with IQR shading for continuous parameters, panels (d)–(e) show bar charts for categorical parameters, and panel (f) presents the 2D heatmap. The default configuration ($d_H = 6$, $w = 15$) lies in a broad plateau rather than at an isolated optimum, and no region of the grid falls below $d = 0.18$.

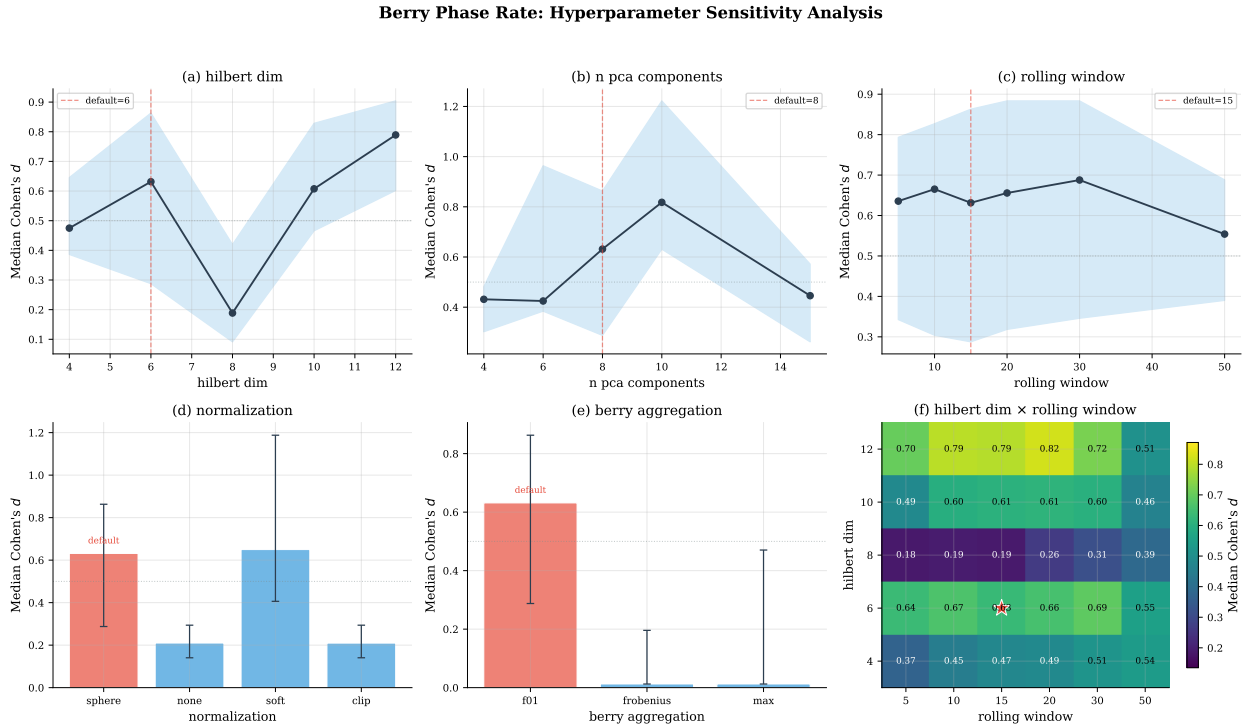


Figure 10: Berry Phase Rate hyperparameter sensitivity. (a)–(c) One-at-a-time sweeps for continuous parameters; vertical dashed line marks the default, horizontal dotted line marks $d = 0.5$. (d)–(e) Categorical parameters; default bar highlighted. (f) 2D grid over Hilbert dimension and rolling window; star marks default (6, 15). Berry Phase Rate maintains $d > 0.5$ across most settings.

I Detection-Oriented Metrics

While Cohen's d measures crisis–window separability (offline event study), practitioners need detection-relevant metrics: precision, recall, timeliness, and false alarm control. We evaluate each method's

z-score series using a threshold of $z > 2$ and three tolerance windows (5, 10, 30 trading days). Table 19 reports aggregates across all 14 crises.

Table 19: Detection metrics across 14 crises (threshold $z > 2$, tolerance window = 10 trading days). Med. Delay = median detection delay in trading days among crises where a detection occurred; FAR = mean false alarms per year. AUC-PR = area under precision–recall curve (threshold-free). “—” = no crisis detected.

Method	F1@10d	AUC-PR	Med. Delay	FAR/yr	Category
Geo. Consensus	0.004	0.024	19.5	3.1	Geometric
Rolling Vol Z	0.004	0.079	25.0	10.8	Classical
Berry Curv. Incr.	0.003	0.027	8.0	8.3	Geometric
Multi-Lag Fidelity	0.003	0.021	5.0	12.4	Geometric
Isolation Forest	0.002	0.111	8.5	17.8	Classical
QCML Chern	0.002	0.011	—	13.5	Geometric
Random Forest	0.001	0.090	10.0	15.0	Supervised
CUSUM	0.001	0.053	0.0	186.6	Classical
QFI Determinant	0.000	0.017	26.0	16.9	Geometric
HMM 2-state	0.000	0.330	—	5.4	Classical
BOCPD	0.000	0.018	—	1.0	Classical

F1 scores are low across all methods due to the extreme class imbalance inherent in crisis detection (crisis days $\approx 5\text{--}10\%$ of the sample). AUC-PR, which is threshold-free, provides a more discriminating ranking. HMM achieves the highest AUC-PR (0.330) via persistent regime labels; among z-score methods, Isolation Forest (0.111) and Random Forest (0.090) lead, with Berry (0.027) and Multi-Lag Fidelity (0.021) at moderate levels. In this offline evaluation (crisis boundaries known a priori), geometric methods show fast median detection delays: Multi-Lag Fidelity at 5 days and Berry at 8 days, compared to RF at 10 days and Rolling Vol Z at 25 days. These offline delays should not be confused with prospective detection speed; see Table 10 for walk-forward results with strictly causal thresholds. CUSUM fires immediately (0-day delay) but at prohibitive FAR (186.6/yr).

J Supplementary Statistical Tests

J.1 Bayesian Signed-Rank Test

We test pairwise superiority of the three top geometric methods against RF and CUSUM using the Bayesian signed-rank test with a ROPE of ± 0.1 [7]. Table 20 shows posterior probabilities. Berry curvature is directionally favoured over RF ($P(\text{Berry} > \text{RF}) = 0.61$), and Multi-Lag Fidelity is near-equivocal ($P(\text{MLF} > \text{RF}) = 0.41$). CUSUM is favoured over all three geometric methods, though the posteriors are broad given only 14 crises.

J.2 Bootstrap Rank Confidence Intervals

Table 21 presents bootstrap 95% CIs for mean Friedman ranks ($B = 10,000$) across all 11 methods. Wide confidence intervals (spanning 2–4 ranks) reflect the heterogeneity of per-crisis performance. CUSUM (4.57, CI [3.21, 5.93]) and Berry Curvature Increment (4.79, CI [3.21, 6.43]) share the top tier, followed by Rolling Vol Z (4.57, CI [3.29, 5.86]), Isolation Forest (4.86), and HMM (5.14) in a broad overlapping cluster. Random Forest (5.64, CI [4.07, 7.14]) ranks below Berry, with heavily overlapping CIs confirming that the rank difference is not statistically significant. Only BOCPD (10.86, CI [10.64, 11.00]) is clearly separated at the bottom.

Table 20: Bayesian signed-rank test (ROPE = ± 0.1) on per-crisis Cohen’s d . $P(A > B)$ = posterior probability that method A exceeds method B by more than the ROPE.

Comparison	$P(A > B)$	$P(\text{equiv})$	$P(B > A)$
Berry vs RF	0.611	0.010	0.379
QFI Det vs RF	0.203	0.001	0.796
MLF vs RF	0.414	0.039	0.548
Berry vs CUSUM	0.318	0.002	0.680
QFI Det vs CUSUM	0.060	0.009	0.931
MLF vs CUSUM	0.073	0.000	0.927

Table 21: Bootstrap 95% CIs for mean Friedman rank (10,000 resamples). Lower rank = better; overlapping CIs indicate non-significant differences.

Method	Mean Rank	95% CI
CUSUM	4.57	[3.21, 5.93]
Berry Curv. Incr.	4.79	[3.21, 6.43]
Rolling Vol Z	4.57	[3.29, 5.86]
Isolation Forest	4.86	[3.79, 5.86]
HMM 2-state	5.14	[3.57, 6.79]
QCML Chern	5.43	[4.14, 6.79]
Random Forest	5.64	[4.07, 7.14]
Multi-Lag Fidelity	5.93	[4.50, 7.36]
QFI Determinant	6.36	[4.64, 8.14]
Geo. Consensus	7.86	[6.79, 8.86]
BOCPD	10.86	[10.64, 11.00]

J.3 Critical Difference Diagram

Figure 11 shows the Nemenyi post-hoc critical difference (CD) diagram, where methods connected by a horizontal bar do not differ significantly at $\alpha = 0.05$. The diagram confirms a single large clique spanning ranks 4.6–7.9 (CUSUM through Geometric Consensus), with only BOCPD outside. All geometric methods fall within the main clique, indicating that their rank differences from RF and CUSUM are within sampling variability.

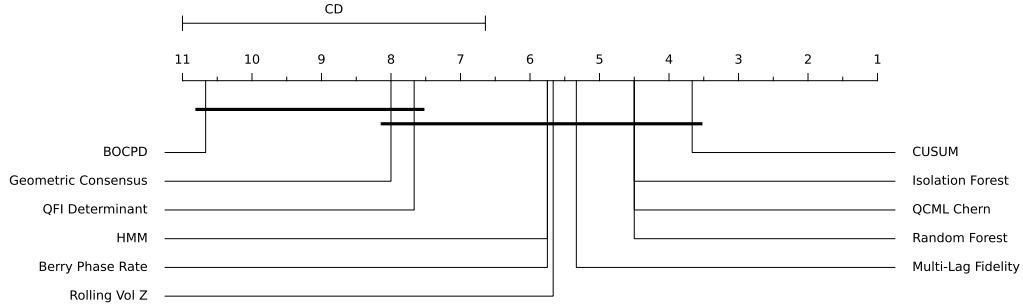


Figure 11: Nemenyi critical difference diagram. Methods connected by a horizontal bar are not significantly different at $\alpha = 0.05$. Axis shows mean Friedman rank (lower = better).

K Hyperparameter Search Spaces

Table 22: Hyperparameter search spaces. Main results use fixed defaults (bold); ablation explores alternatives. Enriched features (4d rolling stats) for all methods.

Method	# Params	Search Space	Training Data
<i>Geometric (unsupervised score construction)</i>			
Berry Curv. Incr.	4	$n \in \{4, 8, 16\}$, $p \in \{10, 15, 20\}$, $op \in \{\text{rnd}, \text{pca}\}$, $w \in \{10, 20, 30\}$	Full (or expanding)
QFI Determinant	4	same as Berry	Full (or expanding)
Multi-Lag Fidelity	4	same as Berry	Full (or expanding)
QCML Chern	3	$n \in \{4, 8\}$, $p \in \{5, 10, 15\}$, $window \in \{10, 20\}$	Full
Geo. Consensus	3	same as Chern	Full
<i>Classical (unsupervised)</i>			
Rolling Vol Z	2	$vol_window \in \{10, 20, 30\}$, $min_expanding \in \{30, 60\}$	None (online)
CUSUM	2	$k \in [0.3, 1.0]$, $burn_in \in \{30, 60, 90\}$	Burn-in period
HMM (2-state)	2	$cov \in \{\text{full}, \text{diag}\}$, $n_iter \in \{50, 100, 200\}$	Full
BOCPD	1	$hazard \in [50, 500]$	None (online)
Isolation Forest	2	$n_est \in [50, 300]$, $contam \in [0.01, 0.15]$	Full
<i>Supervised</i>			
Random Forest	2	$n_est \in [50, 300]$, $depth \in [5, 20]$	LOCO labels
Rolling RF (VIX)	2	$n_est = 200$, $depth = 6$	VIX > 25, 250-day rolling
<i>Oracle</i>			
VIX Level	1	$min_expanding = 60$	None (expanding z-score)

L Operator Ablation

Table 23 compares three operator construction strategies: random Hermitian matrices, n -qubit Pauli tensor products (structured, orthogonal basis for 8×8 Hermitian matrices), and generalized Gell-Mann $SU(N)$ generators (traceless, orthogonal).

Table 23: Operator ablation: mean Cohen’s d across 4 representative crises by construction method.

Observable	Random	Pauli	Gell-Mann
Berry Curv. Incr.	0.82	0.53	0.00
QFI Determinant	0.73	0.31	0.44
Multi-Lag Fidelity	0.43	0.52	0.24

Fixed $n = 8$, $p = 15$, $w = 20$. Pauli: 3-qubit tensor products $\{I, X, Y, Z\}^{\otimes 3}$; Gell-Mann: $SU(8)$ generators. All results use the best scaling exponent for each method (0.0 for Pauli, unscaled for Gell-Mann).

The effect of operator construction is observable-dependent. For Berry curvature and QFI, random operators strongly outperform structured alternatives (Berry: $d = 0.82$ vs. Pauli 0.53; QFI: $d = 0.73$ vs. Pauli 0.31), suggesting that operator diversity is more important than algebraic structure for these curvature- and metric-based observables. Gell-Mann generators kill the Berry signal entirely ($d = 0.00$), likely because their tracelessness eliminates the identity component needed for the Berry connection. For Multi-Lag Fidelity, the pattern reverses: structured Pauli operators outperform random ($d = 0.52$ vs. 0.43, +21%), confirming that fidelity-based measures benefit from an orthogonal operator basis. Table 4 uses the HPO-selected operator per observable: random for Berry, PCA-inspired for QFI and Multi-Lag Fidelity. Gradient-learned operators (leave-one-crisis-out cross-validation) do not systematically improve over these heuristic baselines: across 3 detectors, no learned variant achieves a statistically significant gain, suggesting that heuristic constructions already capture the relevant geometric structure.

M Interaction Test

We classify crises *a priori* as “novel” or “conventional” (Table 3) and methods as “geometric” or “classical.” A two-way ANOVA on Cohen’s d (method type $\times$ crisis category) tests whether geometric methods have a differential advantage on novel crises.

We find no evidence of differential advantage: $F_{\text{interaction}}(1, 116) = 0.38$, $p = 0.54$, partial $\eta^2 = 0.003$. Neither method type ($F = 3.33$, $p = 0.07$) nor crisis category ($F = 3.85$, $p = 0.05$) reaches significance at $\alpha = 0.05$. The stratified finding (Table 8) is thus a *deployment-availability* argument—geometric methods maintain constant performance while RF degrades due to label scarcity—not a claim of differential superiority on novel crises.